



Superconductivity-Induced Shift in Resonant Field in a Superconductor-Ferromagnet Trilayer

Rhiannon Fletcher-Stones

Principal Supervisor: Dr Chiara Ciccarelli (Cavendish Laboratories)

Second Supervisor: Prof Jason Robinson (Materials Science and Metallurgy Department)

July 7, 2025

Abstract

This study explores superconductor-ferromagnet-superconductor (S/F/S) structures using ferromagnetic resonance to elucidate the rigid shift in the resonance field observed with the onset of superconductivity. This shift corresponds to an effective increase in magnetic anisotropy in the ferromagnet facilitated by the superconductor itself. First, prior research is replicated to build a foundation confirming that only S/F/S structures exhibit the rigid shift, not asymmetric structures. This rigid shift is found to be magnetic field-independent, showing nonzero resonance at zero field with a maximal shift of 52 mT. A systematic thickness study reveals that thicker Nb layers amplify the shift. Angular-dependent measurements demonstrate in-plane isotropy of the shift and an enhanced hard-axis anisotropy out-of-plane below the transition temperature. These results suggest that the mechanism is a lensing effect, not of the external field, but from the ferromagnet's magnetisation. Tuneable anisotropy switching could be harnessed for superconducting spintronic and quantum-memory devices.

This report is substantially my own work and conforms to the University of Cambridge's guidelines on plagiarism. Where reference has been made to other research or work done by me in collaboration with others, this is clearly acknowledged in the text, bibliography and Acknowledgements.

Acknowledgments

First and most importantly, I would like to express my gratitude to both of my supervisors on this project, Professor Chiara Ciccarelli and Professor Jason Robinson. Professor Chiara Ciccarelli's insightful guidance and constructive feedback have been instrumental in shaping this project. Her expertise and rigorous approach have taught me to strive for clarity and precision in my work, and I am deeply thankful for the many discussions that challenged and inspired me.

Secondly, I would like to thank Professor Jason Robinson, whose kindness and unwavering support have made this project not only possible but also rewarding. His thoughtful advice and readiness to help me at every stage provided both reassurance and motivation. I have greatly appreciated his perspective and genuine care for my development.

I would also like to thank the Ciccarelli group, and specifically Basit Muhammad, who has been an amazing mentor and friend. Without his help, this project would not have been possible. He was always willing to help with any challenges I faced. Thank you for always listening and being there when things were difficult. I would also like to thank Isaac Wu and Dr. Pankaj Pathak for their advice and support throughout. Finally, I am grateful to the Robinson group for their advice and support on the growth side of the project. They always made me feel welcome and offered valuable guidance.

Contents

1	Introduction	4
2	Background	5
2.1	Magnetism	5
2.1.1	Interactions and Magnetic Energy	6
2.2	Shape Anisotropy	8
2.3	Magnetisation Dynamics	9
2.3.1	Ferromagnetic Resonance	9
2.4	Superconductivity	12
2.4.1	London's Equation	13
2.4.2	Ginzburg Landau Theory	13
2.4.3	BCS Theory	17
2.5	Proximity effects	18
2.5.1	Superconductor/Normal Metal	19
2.5.2	Ferromagnet/Superconductor	19
2.5.3	Long-Range Proximity Effect	20
3	Experimental	21
3.1	Sample Fabrication	21
3.2	Ferromagnetic Resonance Experiment	22
4	Superconductivity induced shift in H_r	24
4.1	Asymmetric Structure	27
4.2	Thickness Dependence	27
5	Angular Dependence	29
5.1	In-Plane	29
5.2	Out-of-Plane	30
6	Discussion	32
7	Future work	33
7.1	Part 1: Finalising Understanding	34
7.2	Part 2: Superconductor-Driven Reorientation of PMA Ferromagnets	35
7.3	Part 3: Inverted Structure	36
7.4	Part 4: Develop a Memory Device	36
7.5	Gantt Chart	36
8	Conclusion	37

1 Introduction

The coexistence of ferromagnetism and s-wave superconductivity has long been thought to be mutually exclusive due to their opposing orderings [1]. Ferromagnetism has symmetric spin states, therefore generally a net magnetisation. Superconductivity is mediated by Cooper pairs (CPs) which are comprised of two electrons with equal and opposite momenta, bound together through lattice interaction, phonons. At the interface of these two materials, we see a destructive interference due to the opposing ordering [2]. However, it has been shown that hybridising superconductivity and ferromagnetism intriguing physics emerges and in recent years extensive research has been undertaken to understand this emergent phenomenon and lead to new device functionality [3]. The main aim of the field of superconducting spintronics is to manipulate the spin state of magnetic materials utilising proximised superconductors. These proximised systems have huge potential in fields such as quantum computation since they could provide a low temperature low energy solution to quantum memory storage. Therefore, superconducting spintronics offers a pathway towards energy-efficient quantum computing which has not been realised [4].

Over the past few decades ferromagnetic Josephson junctions, superconductor-ferromagnet-superconductor (S/F/S), have been heavily researched due to their unusual properties [5, 6, 7, 1]. Recently, a new phenomenon in S-F-S structures has been discovered where the resonant frequency of the ferromagnetic layer in ferromagnetic resonance studies increases below the transition temperature T_c . This is unexpected and corresponds to an effective increase in anisotropy below T_c . The initial study on this phenomenon by Lai-Lai Li et al. 2018 saw a 60 mT drop in the resonant field (H_r) in a S(Nb)-F($\text{Ni}_{80}\text{Fe}_{20}$)-S(Nb) sample with Nb layer of 100 nm and 20 nm of $\text{Ni}_{80}\text{Fe}_{20}$ (Py) [8]. A shift in the resonant field corresponds with a change in the anisotropy landscape of the ferromagnet, an extra contribution to the anisotropy caused by the onset of superconductivity. Interestingly, this effect was only seen in the trilayer structure and the effect disappears in the asymmetric structure (Nb/Py). They attributed this change in anisotropy to spin transfer torque due to spin triplets induced by the magnetisation dynamics at magnetic resonance [9]. This was further explored by Jeon et al. 2019 observed similar effects but with an enhanced Gilbert damping below T_c due to the onset of superconductivity [10]. In 2019 Golovchanskiy et al. reported two follow-up studies involving a thickness dependent study of the different layers and added an insulating. From this they see that the change in the resonant magnetic field (H_r) is enhanced with thicker Nb layers and thicker $\text{Ni}_{80}\text{Fe}_{20}$ and saw the effect disappear when adding a thin insulating layer. In their second paper, they suggested that it had nothing to do with triplet generation since the only mechanism which is possible is through the precession of a ferromagnet. This is unlikely since the frequency required must be in the range of the depairing energy of the Cooper pairs which is much larger than the frequencies used. They explain that it is due to the manifestation of the Anderson-Higgs mass of magnons causing a shift but this is a dynamical effect [11, 12]. This is still not confirmed or well understood.

This report details an experimental investigation into the increase in anisotropy observed below T_c in S/F/S trilayers. The main question is: what physical mechanism causes the rigid shift in the resonance field of $\text{Ni}_{80}\text{Fe}_{20}$ below T_c . To answer this, we have four specific aims:

- To replicate prior measurements of the rigid shift in resonance field below T_c .
- To characterize the thickness dependence of the superconducting and ferromagnetic layers.
- To measure the angular dependence of the resonance field shift.
- To distinguish between candidate mechanisms using both literature and the new data.

2 Background

This section sets out the theoretical background required to understand the effect that is studied in this report. First, basic magnetism theory is discussed alongside magnetisation dynamics which is integral to understanding ferromagnetic resonance. Then basic superconductivity theory is outlined including London's theory, the Meissner effect, Bardeen–Cooper–Schrieffer (BCS) and Ginzburg-Landau theory, the proximity effect is discussed bringing both superconductivity and magnetism together to understand what happens at the interface of these two phenomena.

2.1 Magnetism

Magnetism in materials fundamentally arises from the behaviour of electrons, and it is quantified via the magnetic moment, which is a sum of the angular momentum and intrinsic spin momentum of the system. These magnetic moments interact, and magnetic ordering can emerge as a result. The sum of the magnetic moments in a material is given by the magnetisation M ; therefore, the total field in a material is given by equation 1 where B is the magnetic flux density, H is the magnetic field, μ_0 is the Bohr magneton which is the magnetic moment of a single electron, and χ is the magnetic susceptibility. This is only for the special case when the magnetisation varies linearly with field, and which not always the case [13].

$$B = \mu_0 H + \chi H \tag{1}$$

Magnetic susceptibility describes how a material responds to a magnetic field and is directly related to its electron orbital configurations. In most materials, the electrons are paired up, meaning their moments cancel, and an external magnetic field induces circulating currents which oppose the external field; this is diamagnetic behaviour with negative susceptibility. Superconductors are examples of perfect diamagnets. When the material has uncompensated electrons in the shells, they are forced into the direction of the external field. This is paramagnetic behaviour with positive susceptibility. Without the magnetic field, the material loses all magnetisation. This is due to weak interactions between the spins of the unpaired electrons. If this interaction is strong, we get ferromagnetism, where there is a net magnetisation without an external field. This phenomenon can be observed in transition

metal elements, such as iron (Fe), Nickel (Ni), and Cobalt (Co). This spontaneous magnetisation is only seen below the Curie temperature, where the interaction between spins is stronger than thermal fluctuations [13, 14].

2.1.1 Interactions and Magnetic Energy

To describe the coupling of electron spins, we consider the contributions to magnetic energy. Total free energy in a ferromagnetic system generally results from contributions of exchange energy, magnetostatic energy, Zeeman energy, magnetic anisotropy energy and magnetostrictive energy. These contributions explain how the spins communicate with each other in a material to produce long-range order [15].

Exchange Energy Exchange lies at the very heart of magnetic phenomena. Say we have an electron in state $\psi_a(r_1)$ and an electron in state $\psi_b(r_2)$ their total wave function is a product of this $\psi_a(r_1)\psi_b(r_2)$. This does not obey exchange symmetry since switching electrons leads to $\psi_a(r_2)\psi_b(r_1)$, which is not a multiple of the initial wavefunction. Therefore, the only valid states are symmetrised or antisymmetrised product states. Since the electrons are fermions, they must have an overall wave function that is antisymmetric; therefore, either the spin state must be antisymmetric, χ_S , with a symmetric spatial wavefunction, $S=0$; this is a singlet state. Or we have the symmetric spin state χ_T with antisymmetric spatial state, $S=1$; this is the triplet state [16, 17]. These singlet Ψ_S and triplet state Ψ_T are shown in equations 2 and 3.

$$\Psi_S = \frac{1}{\sqrt{2}} [\psi_a(r_1)\psi_b(r_2) + \psi_a(r_2)\psi_b(r_1)] \chi_S \quad (2)$$

$$\Psi_T = \frac{1}{\sqrt{2}} [\psi_a(r_1)\psi_b(r_2) - \psi_a(r_2)\psi_b(r_1)] \chi_T \quad (3)$$

In the triplet state, the electrons are more spatially separated, which minimises Coulomb interaction, making it more favourable. This leads to ferromagnetism. To quantify this interaction, the difference in energy over 2 represents the exchange constant. With this, we can write the spin-dependent Hamiltonian [16].

$$H_{ex} = -(E_S - E_T) \mathbf{S}_1 \cdot \mathbf{S}_2 = -2J_{ex} \mathbf{S}_1 \cdot \mathbf{S}_2 \quad (4)$$

This can be further applied to the whole system by utilising the nearest neighbour approach. The exchange interaction is shown below in equation 5. This is the Heisenberg model [18].

$$H_{ex} = - \sum_{i,j} J_{ij} \mathbf{S}_1 \cdot \mathbf{S}_2 \quad (5)$$

The sign of J_{ij} determines if the material is ferromagnetic or antiferromagnetic. Suppose two electrons are on the same atom. In that case, the exchange interaction is usually positive to ensure antisymmetric space, minimising Coulomb interaction, which makes the triplet state more favourable for a ferromagnet. To determine the exchange energy in a ferromagnetic material, we need to replace

the spins in the equation with continuous magnetisation $m = M/M_s$, where M_s is the saturation magnetisation, and integrate over the whole volume of the sample shown in equation 6 [13].

$$E_{ex} = \frac{2JS^2z}{a} \int_V (\nabla \mathbf{m})^2 dV \quad (6)$$

Extending this model to transition metals, which display itinerant magnetism, can be explained through ferromagnetic band theory. In transition (ferromagnetic) metals, the outer occupied (or semi-occupied) bands are 4s and 3d, with spin-up $3d\uparrow$ and spin-down $3d\downarrow$. These bands are shifted due to the exchange of energy, resulting in long-range order as long as the exchange energy is greater than the thermal energy [19].

The exchange energy determines magnetic order; however, it is not the whole picture as it does not control magnetic inhomogeneity or domain structure. The final magnetic configuration minimises the total free energy density.

Zeeman Energy

The Zeeman energy arises from the interaction between the magnetic moments and the external field. The interaction between the field and the moments causes the energy bands to be split [13]. The energy of this splitting is the Zeeman energy given by equation 7.

$$E_z = -\mu_0 \int_V \mathbf{M} \cdot \mathbf{H}_{ext} dV \quad (7)$$

This is minimised when $\mathbf{M}$ is parallel to H_{ext} and V is the total volume of the sample.

Demagnetisation Energy

So, exchange energy explains why ferromagnetism is energetically favourable, and the Zeeman effect describes how the magnetic field affects the magnetic moments. However, in practice, there is not usually a single magnetic orientation; instead, competition exists between the exchange energy and the energy associated with dipolar fields. Since $\nabla \cdot \mathbf{H} = -\nabla \cdot \mathbf{M}$ then whenever the magnetisation stops and starts, say, at the edges of the sample, the magnetic field diverges, which creates a demagnetising field. There is an energy cost associated with this of $B^2/2\mu_0$ per cubic metre. The energy associated is given by:

$$-\frac{\mu_0}{2} \int_V \mathbf{M} \cdot \mathbf{H}_d d\tau \quad (8)$$

This energy is minimised when the magnetisation is broken into many domains. However, domains also cost energy. Therefore, there is a balance between the energy cost of the demagnetisation field and the energy cost of creating a domain [13, 14].

Anisotropy Energy

In a real ferromagnet, the spontaneous magnetisation does not randomly orient. It aligns along the easy axis; the direction of this axis is determined via the anisotropy of the material. The preferred orientation of the magnetisation is determined by the magnetocrystalline anisotropy energy (MAE), which is the change in the free energy for a crystal upon rotation of the magnetisation. This preferred direction of magnetisation cannot be due to exchange energy since it is isotropic. An interaction is required that couples the spins with the crystalline field, suggesting that the observed anisotropy is due to the interatomic dipole-dipole interaction between the spins, which leads to a macroscopic shape anisotropy. Shape anisotropy leads to the easy axis of a thin film to lie in-plane to reduce the demagnetising energy, however, even in the case of an infinitely extended homogeneous sample without borders. However, this effect is much smaller than what was seen experimentally, which means there must be another contribution [13, 14].

Van Vleck attributed the preferred magnetisation to spin-orbit coupling, which is the coupling between the spin of the electron and the magnetic field created by its orbital motion around the nucleus. The orbital motion is coupled to the lattice via the electric potential of the ions [20]. Two common magnetocrystalline anisotropies considered in this report are the uniaxial and cubic anisotropy. In uniaxial systems such as Co, which has a hexagonal lattice, the anisotropy leads to energy:

$$E = K_1 \sin^2(\theta) + K_2 \sin^4(\theta) \quad (9)$$

Where K_1 and K_2 are anisotropy constants, which is how much energy is required to rotate the magnetisation away from the easy axis, and θ is the angle between the easy axis and the magnetisation direction. For materials such as Fe, which have FCC packing, they have cubic anisotropy, which is in the form:

$$E_{\text{anis}} = K_1(\alpha_1^2\alpha_2^2 + \alpha_2^2\alpha_3^2 + \alpha_3^2\alpha_1^2) + K_2(\alpha_1^2\alpha_2^2\alpha_3^2) \quad (10)$$

Where K_1 and K_2 are the cubic anisotropy constants and $m(x,y,z)$ is the cosine of M to the three principal axes. The sixth-order term K_2 is negligible in most cases. Therefore, in some materials with $K_1 > 0$ (e.g. Fe), the easy axis is along the (100) crystalline direction or equivalent. In contrast, in other materials with $K_1 < 0$ (e.g. Ni), the easy axis is along the (111) direction [21].

2.2 Shape Anisotropy

Shape anisotropy arises from the demagnetising field, which is the internal magnetic field that the ferromagnet generates. The demagnetising field is responsible for the formation of magnetic domains, which are uniform areas of magnetisation which minimise the dipole energy. The demagnetising field can be derived directly from Maxwell's equations. In this report the technique of ferromagnetic resonance is utilised. In order to gain a good resonance response the magnetisation has to be saturated

in one direction. This leads to the demagnetising field having the form:

$$\mathbf{H}_d(\mathbf{r}) = -\frac{M_s}{4\pi} \nabla \left[\oint_{\delta V} \frac{\hat{\mathbf{n}}' \cdot \hat{\mathbf{m}}}{\|\mathbf{r} - \mathbf{r}'\|} dS' \right] \quad (11)$$

where $\mathbf{m} = M_s \hat{\mathbf{m}}$ is saturation magnetisation. In simple geometries, such as those studied in this report, a linear relationship can be approximated between $\mathbf{H}_d = -N\mathbf{M}$, where N is the demagnetisation tensor, which depends on the sample geometry.

This report focuses solely on planar sample geometries, which enable us to simplify N . The samples can be approximated as thin films, which are infinitely wide slabs with a small thickness, which is a limiting case. For this geometry, $N_x = N_y = 0$ and $N_z = 1$ [15].

2.3 Magnetisation Dynamics

Magnetisation dynamics studies how the magnetisation of a material changes with time. Within this report, ferromagnetic resonance is utilised to probe magnetic properties. In the following section, we describe the equation of motion governing the magnetisation dynamics of precessional motion. We also discuss the origin of magnetisation damping.

2.3.1 Ferromagnetic Resonance

As discussed in section 2.1, magnetism is quantified using the magnetic moments. The magnetic moment can be written in the form of angular momentum classically as shown below:

$$\mu = -\frac{e}{2m} \mathbf{L} = \gamma \mathbf{L} \quad (12)$$

Magnetisation dynamics studies how these moments behave over time; therefore, to model these effects, we take the derivative of equation 12.

$$\frac{d\mu}{dt} = \gamma \frac{d\mathbf{L}}{dt} = 0 \quad (13)$$

This is equivalent to Newton's equation of motion where $\tau = \frac{d\mathbf{L}}{dt}$. When the electron is placed in an external field $\mathbf{H}$, the field exerts a torque $\mu \times \mathbf{H}$. We can extend this to a collective system as well by utilising the magnetisation $\mathbf{M} = n\mu$ hence:

$$\frac{1}{\gamma} \frac{d\mathbf{M}}{dt} = \mathbf{M} \times \mathbf{H}. \quad (14)$$

This represents the Landau-Lifshitz equation, which describes the magnetisation dynamics in an external field where γ is the gyromagnetic ratio [22]. This is not the complete picture since, in reality, it is experimentally observed that the magnetisation aligns after a finite time with the direction of minimal energy. To account for this, damping is introduced via the Gilbert damping, α , which models a viscous damping. Therefore, equation 15 models the magnetisation dynamics [23].

$$\frac{d\mathbf{M}}{dt} = -\gamma \mathbf{M} \times \mathbf{H}_{eff} + \frac{\alpha}{M_s} (\mathbf{M} \times \frac{d\mathbf{M}}{dt}), \quad (15)$$

where $\mathbf{H}_{eff}$ is the effective field which includes the applied fields, the demagnetising field, the crystalline anisotropy field and more. The LLG equation is only valid for large damping systems, and we assume the following. The Gilbert damping parameter α is strictly phenomenological and was introduced to mimic experimental results. Since the process of energy dissipation in a ferromagnet can be very complex with multiple contributions, it cannot be fully described with a single damping parameter [15].

Resonance of the quantum mechanical spins can be achieved through microwave excitation. When an external magnetic field $\mathbf{H}$ is applied, the energy levels of the electrons are split due to the Zeeman effect splitting the energy levels. Resonance can be achieved by applying a microwave signal perpendicular to the external field when the energy E of the incident radiation equals the energy difference between Zeeman levels [24]:

$$E = g\mu_B B = h\nu. \quad (16)$$

To drive ferromagnetic resonance, an oscillating magnetic field is needed to match the frequency of the precessional motion. The resonant frequency can be derived directly from the LLG equation. The effective field discussed above is directly derived from the total magnetic energy, given by:

$$\mathbf{H}_{eff} = \frac{\partial E}{V \partial \mathbf{M}}. \quad (17)$$

Utilising the Smit and Beljers resonance formula which allows us to find the resonance condition. It is also possible to find it from the dynamic susceptibility; however, in this report, this version allows anisotropy to be easily added and leads into angle dependence [25]. The resonance condition can be solved for any specific field orientation by solving:

$$\left(\frac{2\pi f}{\mu_0 \gamma} \right) = \frac{1}{V^4 M_s^2 \sin^2(\psi_M)} \left[\frac{\partial^2 E}{\partial \psi_M^2} \frac{\partial^2 E}{\partial \theta_M^2} - \left(\frac{\partial^2 E}{\partial \psi_M \partial \theta_M} \right)^2 \right]. \quad (18)$$

In this thesis, the main energy contributions considered are the Zeeman energy E_z , the demagnetisation energy E_d , and the energy associated with crystalline anisotropy. The coordinate system used in this derivation is shown in figure 1.

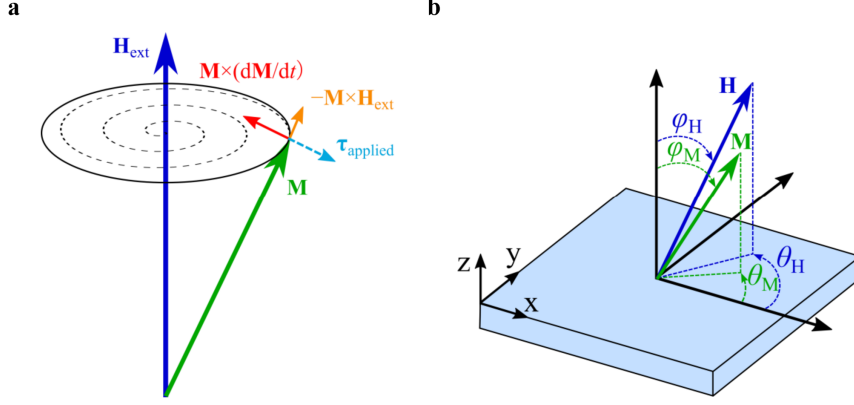


Figure 1: (a) The schematic shows the mechanism for ferromagnetic resonance. (b) The spherical coordinate system utilised in this report where θ is azimuthal angle and φ is the polar angle. This is adapted from [15].

The main energy contributions can be written in spherical coordinates under the coordinate system:

$$E_z = -\mu_0 M_s H [\sin \theta_M \sin \theta_H \cos(\phi_M - \phi_H) + \cos \theta_M \cos \theta_H] \quad (19)$$

$$E_d = \frac{1}{2} \mu_0 M_s^2 (N_x \sin^2 \theta_M \cos^2 \phi_M + N_y \sin^2 \theta_M \sin^2 \phi_M + N_z \cos^2 \theta_M) \quad (20)$$

$$E_a = -\frac{1}{2} M_s \left[H_{2\perp} \cos \psi_M + H_{2\parallel} \sin^2 \psi_M \sin^2 \left(\theta_M = \frac{\pi}{4} \right) + \frac{1}{2} H_{4\parallel} (3 + \cos 4\theta_M) \sin^4 \psi_M \right]. \quad (21)$$

Where $N_{x,y,z}$ are the demagnetisation factors for each axis and $H_{2\perp}$, $H_{2\parallel}$, and $H_{4\parallel}$ are the anisotropy constants which are related to the anisotropy energy via $H_i = \frac{2K_i}{M_s}$. In most experiments, the external field points in a plane, and we assume isotropic anisotropy in the samples studied. This can be greatly simplified by assuming $\psi_M = \psi_H = 90^\circ$ and $\theta_M = \theta_H = \theta$ [26]. If equations 19-21 are inputted into the Smit and Beljers resonance condition 18 we obtain:

$$f = \frac{\mu_0 \gamma}{2\pi} \sqrt{(H_{\text{res}} + H_1)(H_{\text{res}} + H_2)} \quad (22)$$

where

$$H_1 = M_s [-N_x \cos^2(\theta) - N_y \sin^2(\theta) + N_z] - H_{2\perp} + H_{2\parallel} \sin^2 \left(\theta - \frac{\pi}{4} \right) + \frac{1}{4} H_{4\parallel} (3 + \cos 4\theta) + H_U \sin^2 \theta \quad (23)$$

$$H_2 = M_s (N_y - N_x) \cos 2(\theta) - H_{2\parallel} \sin(2\theta) + H_{4\parallel} \cos 4\theta - H_U \cos 2\theta \quad (24)$$

If we take the assumption that the sample is an infinitely thin plane with negligible crystalline anisotropy, so $N_x = N_y = 0$ and $N_z = 1$ and $H_{2\parallel} = H_{4\parallel} = 0$. This leads us to the Kittels formalism for the in-plane case [27].

$$f = \frac{\mu_0 \gamma}{2\pi} \sqrt{H_{\text{res}}(H_{\text{res}} + M_{\text{eff}})} \quad (25)$$

The effective magnetization M_{eff} is given by

$$M_{\text{eff}} = M_s + H_{2\perp} \quad (26)$$

where M_s is saturation magnetisation and $H_{2\perp}$ is the second-order perpendicular anisotropy field, defined as:

$$H_{2\perp} = \begin{cases} > 0, & \text{perpendicular easy-axis anisotropy,} \\ < 0, & \text{in-plane easy-plane anisotropy.} \end{cases}$$

Thus:

- If $H_{2\perp} > 0$, the film has an out-of-plane easy axis and $M_{\text{eff}} > M_s$.
- If $H_{2\perp} < 0$, the film has an in-plane easy plane and

$$M_{\text{eff}} = M_s + H_{2\perp} = M_s - |H_{2\perp}| < M_s.$$

This unified $+H_2^\perp$ convention avoids having to change the sign of the anisotropy term depending on its character [21].

The Gilbert damping can also be determined by measuring the linewidth of the resonance peak as a function of frequency.

$$\Delta H = \Delta H_0 + \frac{2\pi f \alpha}{\mu_0 \gamma} \quad (27)$$

Where ΔH_0 is the inhomogeneous line broadening, which is caused by disorder in the material [15].

2.4 Superconductivity

Superconductivity was discovered in 1911 by H. Kamerlingh Omnes when helium was first liquified. He found that below a characteristic critical temperature (T_c), the electrical resistance drops to zero. Due to the zero-resistance behaviour, persistent currents flow without dissipation for over a year, with an estimated time to decay of 10^5 years. This discovery was very exciting at the time, as the technological opportunity resulting from this effect was huge as was the immense opening of new Physics. Perfect conductivity is the first hallmark of superconductivity and allows us to have high current dissipation lines and strong electromagnets [28].

In modern times, zero resistivity is not considered the definitive proof of superconductivity. Proof requires observation of the Meissner effect. Superconductors expel magnetic fields; this might initially seem to be explained by their perfect conductivity; however, when a superconductor is cooled with a magnetic field, it is expelled at the point of becoming superconducting, which is not explained through perfect conductivity, as conductors trap flux. This effect is more concrete in displaying superconductivity, as it is an equilibrium effect, unlike zero resistance, which depends on transport. To maintain the expulsion of the magnetic field, there must be screening currents on the surface that induce opposing fields. Since the Meissner effect is reversible, it implies that superconductivity is destroyed by a

critical magnetic field H_c , which is related to the difference in free energy between the superconducting state and the normal state, known as the condensation energy [29, 28]. It was found empirically that the temperature dependence is:

$$H_c = H_c(0)[1 - (T/T_c)] \quad (28)$$

2.4.1 London's Equation

The electrodynamical properties are based on these two effects, and the first theory developed from these effects was formulated by the London brothers. Their theory was primarily inspired by the superfluid two-fluid model, which suggests that a fraction of electrons become superfluid (n_s) and the others remain normal, exhibiting finite resistance (n_n), where $n = n_s + n_n$. The superfluid electrons effectively short out the normal electrons, which is what causes the zero-resistance reading. This leads to the London equation, which relates the electrical current in the superconductor [28, 30]. This is derived by considering Ohms Law for AC fields, this is given by: $\mathbf{j}e^{i\omega t} = \sigma(\omega)\mathbf{E}e^{i\omega t}$, then by utilising Faraday's law $\nabla \times \mathbf{E} = -\frac{d\mathbf{B}}{dt}$ and then the Drude model $\sigma(\omega = 0) = \frac{ne^2}{i\omega m}$ we get to the London's equation.

$$\nabla \times \mathbf{j} = -\frac{n_s e^2}{m_e} \omega \mathbf{B} \quad (29)$$

This equation establishes a relationship between $\mathbf{B}$ and $\mathbf{j}$, which is crucial for understanding both the Meissner effect and zero-resistance. Combining this equation with Ampere's Law, we get:

$$\nabla \times \nabla \times \mathbf{B} = -\frac{1}{\lambda^2} \mathbf{B} \quad (30)$$

The term λ has the dimensions of length and is defined as London penetration depth, which is one of the important length scales in superconductivity:

$$\lambda = \left(\frac{m_e}{\mu_0 n_s e^2} \right)^{\frac{1}{2}} \quad (31)$$

The London Penetration depth gives the length over which flux penetrates into the superconductor. By utilising the London and Maxwell's equations, we can see that the superconductor screens the magnetic field, given by:

$$\mathbf{B} = \mathbf{B}(0)e^{-\frac{x}{\lambda_L}}, \quad (32)$$

This explains an exponential decay of the flux in the superconductor and it is a useful quantity when describing superconductors [29].

2.4.2 Ginzburg Landau Theory

The superconducting and normal states are separate thermodynamical phases, just like gas and liquid. Each of these thermodynamic phase transitions is characterised by a characteristic temperature; therefore, we can study superconductivity from a thermodynamic perspective. This leads to the phenomenological Ginzburg Landau theory [28, 30, 31].

In 1937, Landau proposed his theory on second-order phase transitions, which is quantified via the variation of an order parameter. Landau noticed that typical second-order phase transitions involve some change in symmetry of the system. In ferromagnetism, above T_c , we see no magnetic moment, and below T_c , we see magnetic moments. Therefore, the onset of ferromagnetism can be explained by the magnetisation $\mathbf{M}$, which is the order parameter in this case. For a superconductor, Ginzburg and Landau (GL) postulated the order parameter to be represented by ψ , the macroscopic wavefunction which represents the superconducting state, it obeys:

$$\psi = \begin{cases} \psi, & \text{if } T < T_c \\ 0, & \text{if } T > T_c \end{cases} \quad (33)$$

Where $|\psi|^2$ represents the density of Cooper pairs n_s . GL theory describes the macroscopic behaviour of superconductors by expanding free energy as a power series. It operates under two assumptions: that we are close to the transition temperature and that the order parameter varies smoothly.

$$f_s = f_n + \alpha|\psi(r)|^2 + \beta|\psi(r)|^4 + \frac{1}{2m^*} \left| \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right) \psi(r) \right|^2 + \frac{h^2}{8\pi} \quad (34)$$

The first term in the free energy represents the normal state energy, so if $\psi = 0$, we are left with f_n . The second and third terms are the potential terms from the power series expansion where α and β are temperature-dependent phenomenological expansion coefficients. There are only even powers since odd powers introduce a complex component, and free energy must be real and invariant under inversion symmetry. The third term represents the kinetic term from gradients and how the condensate interacts with fields. Finally, we have the magnetic energy. If we first consider the case without fields and gradients, we have to assume that β is positive so that the free energy does not diverge. Therefore, there exist two cases: one with $\alpha > 0$ and one with $\alpha < 0$ shown in figure 2.

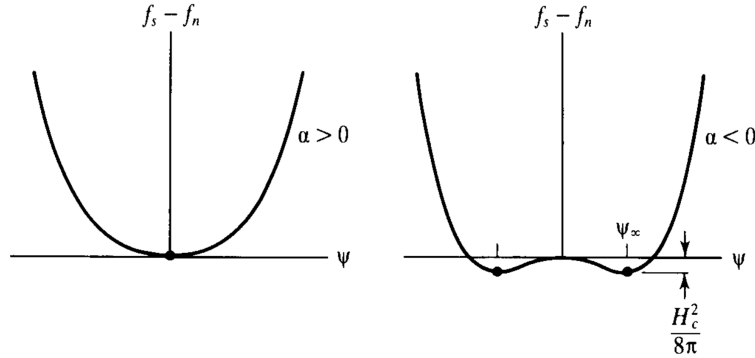


Figure 2: Ginzburg Landau theory free energy in zero magnetic field and without gradients. There are two cases where $\alpha > 0$ in the normal state with minima when the order parameter is zero and $\alpha < 0$ in the superconducting state where it is favorable to have a condensate [29]

From figure 2, we can see two distinct cases. The first shows the normal state, which has free energy minimised when $\psi = 0$. However, when $\alpha < 0$, a lower energy state opens up, corresponding

to the superconducting state. By minimising free energy, we can find the equilibrium point to be $|\psi(r)|^2 = -\frac{\alpha}{\beta}$. Since α changes T_c , we can Taylor expand to leading order in T around T_c to find that [28].

$$\alpha(T) = \alpha'(T - T_c) \quad (35)$$

Therefore, we can write the order parameter as

$$|\psi| = \sqrt{\frac{a'}{b}} \sqrt{T_c - T} \quad (36)$$

This illustrates the temperature dependence of the order parameter, as shown in the plot.

Now that we have considered the case with no fields and gradients, we can examine everything together. The order parameter $|\psi| = \psi e^{i\theta}$ adjusts itself to minimise free energy. Through variational methods, we end up with the GL differential equations.

$$\alpha\psi + \beta|\psi|^2\psi + \frac{1}{2m^*} \left(\frac{\hbar}{i} \nabla - \frac{e^*}{c} \mathbf{A} \right)^2 \psi = 0 \quad (37)$$

$$\mathbf{j}_s = \frac{e^* \hbar}{2im^*} (\psi^* \nabla \psi - \psi \nabla \psi^*) - \frac{e^{*2}}{m^* c} |\psi|^2 \mathbf{A} \quad (38)$$

The first GL equation describes how the order parameter varies in the presence of electromagnetic fields. It describes how the superconducting state varies under spatial changes or fields. The second GL equation describes the supercurrent in terms of the order parameter and the magnetic vector potential, and it describes how the supercurrent reacts to phase variations and electromagnetic fields.

To get a feel for the GL equations, we can simplify the first equation by setting $\mathbf{A} = 0$ and introduce $f = \frac{\psi}{\psi_\infty}$ where ψ_∞ is the value of the order parameter deep inside the superconductor and acts as a normalisation factor.

$$\frac{\hbar}{2m^*|\alpha|} \frac{d^2 f}{dx^2} + f - f^3 = 0 \quad (39)$$

Equation 39 is solved with an exponential decay over a characteristic length scale, which is called the coherence length $\xi(T)$, which describes how superconductivity dies down at boundaries and how quickly it recovers at defects. This is the second important length scale needed to understand superconductivity.

$$\xi(T) = \frac{\hbar^2}{2m^*|\alpha(T)|} \quad (40)$$

Where GL theory is particularly strong is in describing inhomogeneous systems since microscopic theories cannot. There are two key types of superconductors. Type I is the classic superconductor, characterised by a single sharp transition at T_c , exhibiting perfect diamagnetism. Type II superconductors, on the other hand, exhibit a gradual transition. At H_{c1} , they enter a mixed state of superconducting and normal state coexist since it is energetically favourable to allow flux to penetrate. Then, at H_{c2} , the superconductor transitions to the normal state. GL theory enables us to

study type II superconductors, as it accounts for spatial variations in the superconducting state. This can be seen in figure 3.

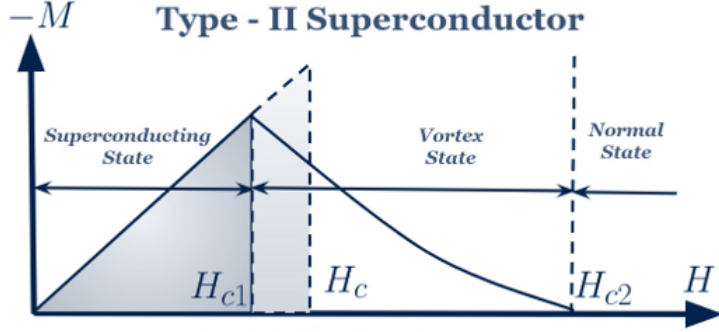


Figure 3: Type I vs type II superconductor showing how the superconductor responds to a external magnetic field. The key difference is the type II lets flux enter creating a vortex state therefore having two critical fields [32].

So far in this thesis, two length scales have been introduced to quantify the difference between Type I and Type II superconductivity [30]. Therefore, we introduce the parameter:

$$\kappa = \frac{\lambda(T)}{\xi(T)} \quad (41)$$

The classification of superconductors based on κ :

Type I Superconductors	Type II Superconductors
$\xi(T) > \lambda(T) \quad \sigma > 0$ The order parameter extends over a greater distance, and the magnetic field does not penetrate.	$\xi(T) < \lambda(T) \quad \sigma < 0$ Magnetic fields penetrate more easily compared to the coherence length.

This means that type II superconductors' energy is reduced when the field is allowed to penetrate, making boundaries favourable due to the longer penetration depth. Due to flux quantisation, the flux penetrates in the form of quantised vortices, with a diameter on the scale of the coherence length, in a hexagonal lattice known as the Abrikosov lattice. Physically, the characteristic length scales are dependent on the material's structure. The coherence length is dependent on the electron-phonon coupling; therefore, a larger electron-phonon coupling causes a longer coherence length, as the coupling between Cooper pairs is stronger. The London penetration depth is dependent on carrier density, and materials with lower carrier density tend to have a larger penetration depth since there is less screening [28].

2.4.3 BCS Theory

GL theory is incredibly powerful; however, the physical meaning of the order parameter was not fully understood until the microscopic Bardeen-Cooper-Schrieffer (BCS) theory was developed. It was not until Gorkov derived GL theory from BCS theory that the physical interpretation of the order parameter was established [28, 2].

The key insight from BCS theory is that there is an effective attraction between electrons, which is counterintuitive to assume due to their repulsive nature due to Coulomb interaction. Frohlich suggested that the attraction could be due to phonon interaction. One electron distorts the positive ions in the lattice, creating a region of slightly higher positive charge. Another electron feels this attraction; the interaction is mediated via phonon exchange and results in a net attractive force. The interaction only happens at the Fermi surface as the electrons cannot scatter deep into the Fermi sea. This effect causes a bound state between two electrons to form called a Cooper pair even if the positive attraction is weak; therefore, the Fermi liquid is unstable.

If we assume a spherical Fermi surface where all states below k_F are occupied and place another two electrons just outside the Fermi surface, they interact via electron-phonon interaction, the two-particle wave function is shown as

$$\Psi(r_1, \sigma_1, r_2, \sigma_2) = e^{i\mathbf{k}_{cm} \cdot \mathbf{R}_{cm}} \varphi(r_1 - r_2) \phi_{\sigma_1, \sigma_2}^{spin} \quad (42)$$

Where $\mathbf{R}_{cm} = (r_1 + r_2)/2$ is the total centre of mass momentum of the system. It turns out that the minimum energy state is when the centre of mass momentum remains unchanged after the interaction. This means that Cooper pairs are formed via electrons with opposite momenta, which is an important realisation.

It was realised that as soon as there was a small effective positive attraction, the whole Fermi surface was unstable, and each electron would be bound to a Cooper pair. This extends the problem to a many-particle problem, forming a coherent state. When Cooper pairs form, they behave like bosons and condense into the ground state due to Bose-Einstein condensation, which creates a lower energy state known as a quantum coherent state. To describe this new quantum coherence, BCS proposed a variational wave function that encapsulates the quantum coherence in the superconducting state.

$$|\Psi_{BCS}\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger \right) |0\rangle \quad (43)$$

Where $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are variational parameters satisfying $|u_{\mathbf{k}}|^2 + |v_{\mathbf{k}}|^2 = 1$, and $c_{\mathbf{k}\sigma}^\dagger$ are the usual fermionic creation operators. One of the most profound outcomes of the BCS framework is the appearance of an energy gap in the excitation spectrum of the system.

The formation of this new BCS ground state causes an energy gap to form between the ground

state and the excited states where no electrons can exist. Excitations are no longer free electrons; they are Bogoliubov quasiparticles, which are mixtures of electrons and holes. In the superconducting state, quasi-particles cannot be excited unless the energy is greater than the gap parameter Δ , which is the energy required to break up a Cooper pair. The quasiparticle dispersion relation is given by:

$$E_k = \sqrt{(\epsilon_k - \mu)^2 + \Delta^2} \quad (44)$$

At finite temperatures the gap $\Delta(T)$ decreases with increasing temperatures:

$$\Delta(T) = \Delta(0) \sqrt{\left(1 - \frac{T}{T_c}\right)} \quad (45)$$

This is a key feature of superconductivity that the gap vanishes continuously at the transition temperature, and the system transitions into the normal state.

2.5 Proximity effects

The next question to answer is how superconductivity reacts when placed next to a normal metal or ferromagnet. With the advent of BCS theory, it became clear that superconductivity and magnetism should be two mutually exclusive effects, as singlet superconductivity has an antisymmetric spin, and ferromagnetism has a symmetric spin. Therefore, the exchange field of the ferromagnet causes the Cooper pairs to break, destroying superconductivity; this is called the paramagnetic effect [33].

Paramagnetic Effect

The paramagnetic effect explains why an external magnetic field breaks the Cooper pairs. When the energy gain from spin polarisation in the normal state $\Delta E_n = -\chi_n H^2/2$ is equal to the condensation energy, the Cooper pairs $\Delta E_s = -N(0)\Delta_0^2/2$ there is a phase transition from the superconducting state to the normal state which happens at:

$$H = \frac{\Delta_0}{\sqrt{2}\mu_B} \quad (46)$$

This is the Chandrasekhar-Clogston limit. Note that this represents the field of a first-order phase transition; the full treatment shows that above a specific temperature, T^* , the phase transition becomes second-order due to thermal fluctuations dominating rather than the paramagnetic effect.

FFLO State

At very low temperatures, when we have high magnetic fields, causing Zeeman splitting but large condensation energy. The Zeeman splitting causes instability in pairing as the Fermi surfaces for a spin up and down shift; therefore, $k \uparrow$ does not equal $k \downarrow$. Now, uniform pairing is not energetically favourable, but it also costs energy to go to the normal state. Therefore, to resolve this, Cooper pairs

will form outside of zero centre of mass ($k \uparrow, -k \downarrow + q$). This causes a spatially modulated superconducting state since the Cooper pairs have a net momentum, so Δ oscillates in space. The modulated order parameter enables the system to partially align with the spin-polarised Fermi surfaces partially, thereby reducing the energy cost imposed by the Zeeman effect. This effect is challenging to observe experimentally, as it occurs at $T = 0.56T_c$ and at high magnetic fields but not high enough to cause the instantaneous destruction of the superconducting state [33].

2.5.1 Superconductor/Normal Metal

When a superconductor is interfaced with a normal metal, the superconductor modifies the material properties of the normal metal near the interface. The Cooper pairs can penetrate into the normal metal by a distance ξ_N in the dirty limit:

$$\xi_N = \sqrt{\frac{\hbar D_N}{k_B T}}. \quad (47)$$

Where D_N is the diffusion constant of the normal metal, $\hbar$ is reduced Plank constant and k_B is the Boltzman constant. In normal metals like Copper, at low temperatures, this distance can extend to a micrometre. On the other hand, at the interface, the superconductor order parameter is depleted on the order of the coherence length ξ_S , the same one derived from GL theory. The phenomenon of induced superconductivity in a normal metal is called the proximity effect, and the effect on a superconductor is called the inverse proximity effect. This is explained through the concept of Andreev's reflection.

Andreev (1964) demonstrated how single-electron states of the normal metal are converted into bound states and also explained the transformation of dissipative electrical current into a dissipationless supercurrent at the interface. When an electron from the normal metal with energy less than the gap encounters a barrier since there are no available states in the superconductor. Therefore, the electron cannot traverse into the superconductor without contributing to the condensate itself. To maintain momentum and charge conservation, a hole is reflected, and another electron is pulled out of the Fermi sea to become a Cooper pair with the initial momentum since the pairing field is strong at the interface. This interaction governs charge transport in superconductor normal metal systems.

2.5.2 Ferromagnet/Superconductor

It is expected that the exchange field of the ferromagnet would destroy the superconducting state at the interface. However, the interaction is much more complex. The physics of the variation of the order parameter was similar to the previously described FFLO state. Firstly, since the spin bands of the ferromagnet are not equally populated, Andreev reflection is suppressed. Secondly, the phase coherence is broken between the electrons and the reflected holes due to the exchange field. For a diffuse ferromagnet, the characteristic penetration depth is on the order of ξ_F , which is often only a few nanometers. For a fully spin-polarised material, the singlet Cooper pairs cannot be formed since

there is only one spin band filled. Thus, the interface becomes completely reflective.

However, there is another peculiar behaviour. Cooper pairs are in a spin singlet state in order to have the symmetric space wave function:

$$|\Psi_s\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \quad (48)$$

However, there are symmetric spin states called triplet states:

$$|\Psi_t^{(+1)}\rangle = |\uparrow\uparrow\rangle, \quad |\Psi_t^{(0)}\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |\Psi_t^{(-1)}\rangle = |\downarrow\downarrow\rangle \quad (49)$$

Since the ferromagnet has an exchange field with spin-dependent band structure, in ferromagnets, there is a strong exchange field which splits the energy levels, meaning if a Cooper pair enters a superconductor, the electrons with momentum $k \uparrow$ and $-k \downarrow$, which form the Cooper pair no longer have the same energy, meaning they must adjust their momentum to be able to sit in the same energy state. In order to pair and still satisfy energy conservation, the electrons must adjust their momenta such that the centre-of-mass momentum of the pair becomes finite, $Q = k \uparrow + k \downarrow \neq 0$. The finite COM momentum causes a spatial modulation of the superconducting pair amplitude by a factor $e^{\pm i(k \uparrow + k \downarrow) \cdot R}$ respectively. As a result, we have:

$$|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \Rightarrow |\uparrow\downarrow\rangle e^{iQ \cdot R} - |\downarrow\uparrow\rangle e^{-iQ \cdot R} = |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \cos(Q \cdot R) + i|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle \sin(Q \cdot R) \quad (50)$$

Thus, it is clear that the singlet and triplet components mix and the overall behaviour is oscillatory.

2.5.3 Long-Range Proximity Effect

The oscillatory behaviour is only short-ranged since the coherence length is only a few nanometers. However, long-ranged proximity effects can be induced by inducing equal spin triplets with $m_s = 1$ and $m_s = -1$. These Cooper pairs with spin triplets are immune to the pair-breaking mechanism of the exchange field. This is possible for even singlet superconductors if there is some magnetic inhomogeneity at the interface or spin-orbit coupling at the interface. Essentially, if the interface is spin-active, it can induce equal-spin Cooper pairs. Therefore, if the magnetisation vector, so the direction of the quantisation, changes direction along the path crossed by the Cooper pairs, this component is rotated in the spin space. As the Cooper pair moves through regions of differing magnetisation directions, its spin state experiences a rotation in spin space. This is maximised when the spin directions are at 90 degrees. The main limiting factor for the propagation of triplet Cooper pairs is now the spin diffusion length, which is a characteristic distance an electron travels before undergoing scattering that results in a spin flip. Once the spin of one of the two electrons is flipped, the triplet alignment is lost, and the exchange interaction breaks the Cooper pairing. This mechanism for generating spin triplets was proposed by Bergeret et al. in their theoretical works.

However, how can triplets exist in a Cooper pair since that would violate the Pauli exclusion principle?

This means that the total wave function has to be antisymmetric under the exchange of two electrons. Equal-spin triplets have a symmetric spin wavefunction; therefore, the orbital must be odd in parity, for example, a p-wave or even an f-wave. This system is not compatible with S/F structures, as the p-wave is highly anisotropic and would not support the long-range proximity effect. This suggests that triplet coupling is impossible in these systems. However, if we consider the time dependence of the wave function and if this is odd in frequency (time), there is a temporal separation between the two electrons. They do not occupy the same state at the same time. This only occurs in S/F systems with broken spin-rotation symmetry, as it generates odd-frequency components [34, 35].

3 Experimental

This section of the report outlines the fabrication and experimental procedures required to measure the superconductivity-driven shift in the resonant field. Samples were grown using DC magnetron sputtering, and field-modulated ferromagnetic resonance (FMR) was the primary measurement technique used.

3.1 Sample Fabrication

The samples studied in this investigation are grown by DC magnetron sputtering. This is a physical vapour deposition technique that begins by introducing an inert gas, such as argon, into an ultra-high vacuum chamber where high voltages are applied between the target and an anode, which ionises the argon gas, creating a plasma. However, in magnetron sputtering, a magnetic field is applied behind the target, which effectively traps electrons near the target's surface, increasing the local electron density and enhancing the plasma density near the surface, thereby improving the sputtering efficiency. The ionised argon plasma is attracted towards the cathode, which serves as the target, causing the ions to collide with the target atoms and eject them. This is the process of sputtering. The sputtered atoms travel through the vacuum chamber, where they reach the substrate and condense on the surface, growing a thin film over time. The properties of the film can be adjusted by changing argon pressure, temperature, deposition time and more sputtering parameters [36].

The samples under study in this work are S/F/S structures, specifically Niobium (Nb)/Permalloy ($\text{Ni}_{80}\text{Fe}_{20}$)/Niobium (Nb) thin films with varying thicknesses. They are grown on Si/SiO₂ substrates, which are cut into 5×5 mm squares and prepared by sonication in acetone for 2 minutes, then sonication in IPA for 2 minutes. The $\text{Ni}_{80}\text{Fe}_{20}$ was grown at 60W and 3 mTorr which has a growth rate of 0.031 nm/s. The deposition rate was characterised by atomic force microscopy. The Nb was grown at 30 W with two Nb guns and 4 mTorr, with a growth rate of 0.072 nm/s. The growth rate was previously determined through the same method explained above by the Robinson group.

The structures investigated in this report are shown in table 1. Six samples were investigated with varying Nb and $\text{Ni}_{80}\text{Fe}_{20}$ thickness, and also studied an asymmetric structure, which is the bilayer

sample S6. The error in growth is roughly 3% for atomic force microscopy calibration.

Sample	Nb thickness (nm)	Ni ₈₀ Fe ₂₀ thickness (nm)	Nb thickness (nm)
S1	110 $\pm$ 3.3	20 $\pm$ 0.6	110 $\pm$ 3.3
S2	110 $\pm$ 3.3	50 $\pm$ 1.5	110 $\pm$ 3.3
S3	110 $\pm$ 3.3	10 $\pm$ 0.3	110 $\pm$ 3.3
S4	80 $\pm$ 2	20 $\pm$ 0.6	80 $\pm$ 2.4
S5	50 $\pm$ 2	20 $\pm$ 0.6	50 $\pm$ 1.5
S6	110 $\pm$ 3	20 $\pm$ 0.6	0

Table 1: Samples of study

3.2 Ferromagnetic Resonance Experiment

The experimental setup is housed inside an Oxford Instruments Teslatron cryostat, which can reach a base temperature of 1.5 K [37]. The cryostat is displayed in figure 4.

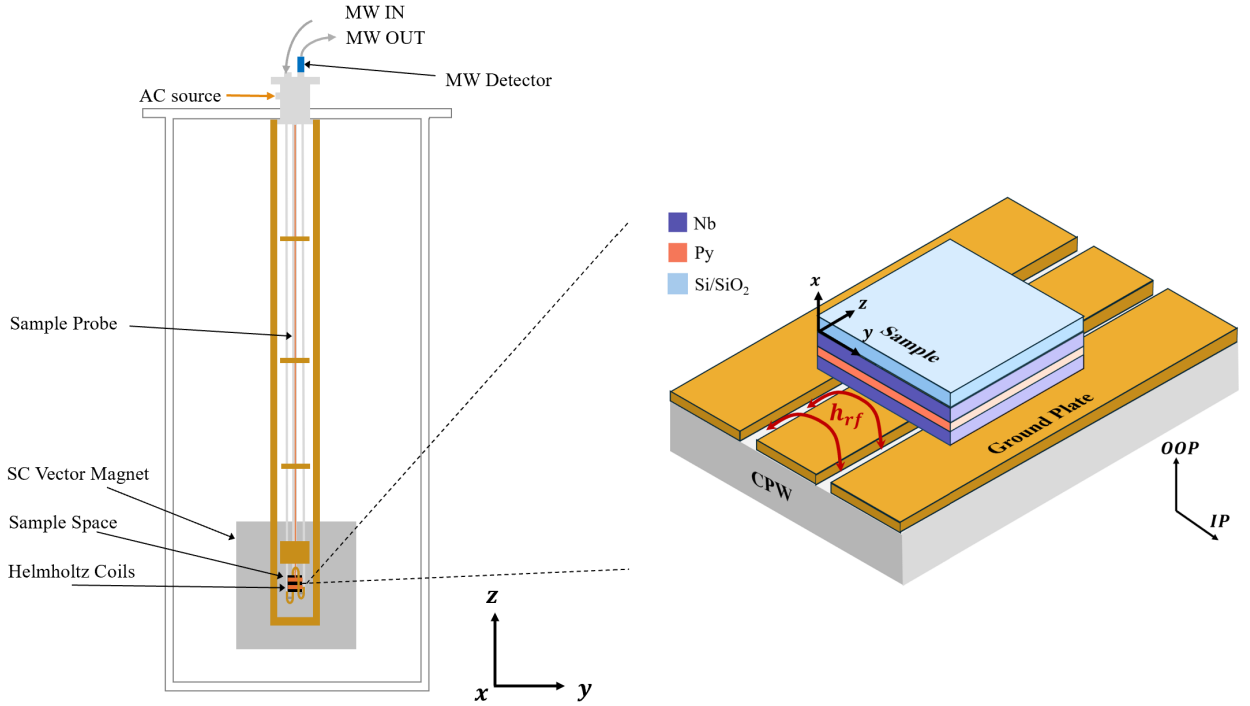


Figure 4: Schematic diagram of the 1.5 K cryostat and probe assembly. The cryostat is equipped with a three-axis superconducting vector magnet (6 T along the primary axis z , 1 T in each of the two orthogonal axes, x and y). The low-temperature probe houses a coplanar waveguide (CPW) for microwave excitation and detection, together with a modular sample mount and Helmholtz coil, allowing precise modulation of the external field.

Cryostat Environment

The sample is mounted using vacuum grease onto a coplanar waveguide (CPW), and placed into the cryostat. The external magnetic field and cryogenic environment are provided by the Oxford Instruments Teslatron cryostat, which contains a superconducting vector magnet with 6 T in the z-direction, 2 T in the x- and y-directions, and a base temperature of 1.5K. The sample is mounted on a sample probe, where the thickness lies in the x-direction, and is designed so that the sample is positioned between the superconducting coils to maximise the uniformity of the field. Steel coaxial cables run down the length of the sample probe and act as microwave transmission lines to the CPW. Connected to the top of the cryostat, there are copper coaxial cables. The first copper cable acts as the input and is connected to a microwave generator, which drives the AC field that excites the sample. The second cable is connected to a microwave detector, which is directly connected to a lock-in amplifier.

Field Modulation

The FMR is field-modulated to enhance sensitivity, as the FMR signal is very small. This is done through Helmholtz coils, which are mounted on the sample probe. The Helmholtz coils are connected directly to a universal source, which provides a modulation current of 0.1 mA. The FMR is field modulated to ensure the uniformity of the field and also reduces noise, as the lock-in uses the modulation frequency, which is set at 321 Hz, as a reference, allowing it to filter out all other frequencies. 321 Hz is chosen since it avoids the mains frequency of 50 Hz and any higher harmonics [38].

Field Sweep FMR

Ferromagnetic resonance can be conducted by either sweeping frequency or magnetic field. In this case, the FMR absorption spectrum is detected using a microwave power detector, typically over a frequency range of 2-18GHz. At fixed frequencies, the magnetic field is swept to find the resonance field. The resulting data is in the form of the derivative of the Lorentzian, which is analysed using the symmetric and antisymmetric Lorentzian functions with respect to H_{res} . This is fitted with the following expression:

$$\frac{dV}{dH} = S \left(\frac{(\Delta H)^2(H - H_{res})}{[(\Delta H)^2 + (H - H_{res})^2]^2} \right) + A \left(\frac{(\Delta H)(H - H_{res})^2 - (\Delta H)^3}{[(\Delta H)^2 + (H - H_{res})^2]^2} \right) + aH + d, \quad (51)$$

where H is the applied external field, H_{res} is the resonance field at which maximum absorption occurs, and ΔH is the linewidth of the resonance peak. The coefficients S and A correspond to amplitudes of the symmetric (absorptive) and antisymmetric (dispersive) components of the Lorentzian lineshape, respectively. The terms aH and d represent a linear background and a constant offset, which account for non-resonant contributions to the voltage signal. The resulting lineshape is shown in figure 5.

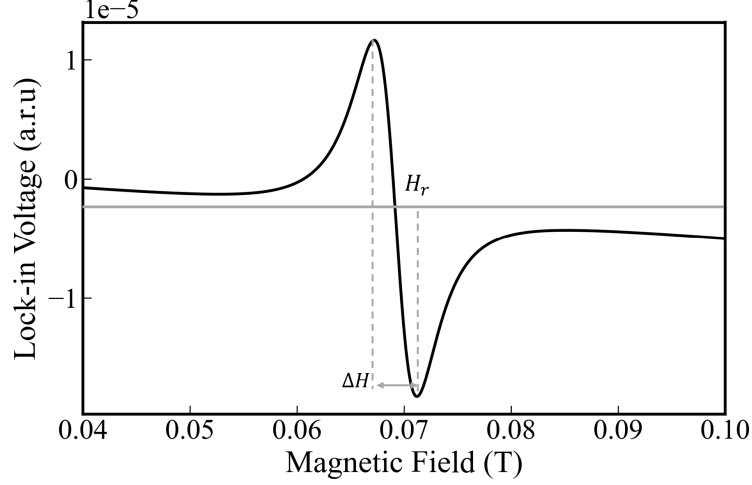


Figure 5: Output Lorentzian measured through lock-in techniques. The resonant field is extracted in the centre of the Lorentzian. The linewidth is found from the difference in position of the two lobes.

From this, the resonant field and linewidths can be extracted from each scan at each frequency to form the Kittel relationship, allowing for exploration of changes in the frequency-magnetic field relationship. The damping parameters, such as Gilbert damping α and inhomogeneous line broadening ΔH_0 , can be extracted by plotting a frequency against a linewidth plot, which gives a linear relationship. The gradient of this allows the Gilbert damping to be extracted, and the inhomogeneous broadening is extracted from the intercept [27].

4 Superconductivity induced shift in H_r

The first investigation aimed to replicate the results from previous literature on this effect [8, 11, 12, 10, 39]. Firstly, to observe the shift in H_r and to confirm that this is only seen in the trilayer structure, not the bilayer. Then, a thickness study was completed on six different samples to determine the effect of varying the thickness of both Nb and $\text{Ni}_{80}\text{Fe}_{20}$ has on this effective shift. All measurements were performed from 10K to 1.5K over a frequency range of 6 GHz to 15 GHz, utilising field modulation FMR discussed in section 3.2. Figure 6 illustrates the studied phenomenon of the S4 sample which is in this structure Nb(80 nm)/ $\text{Ni}_{80}\text{Fe}_{20}$ (20 nm)/Nb(80 nm).

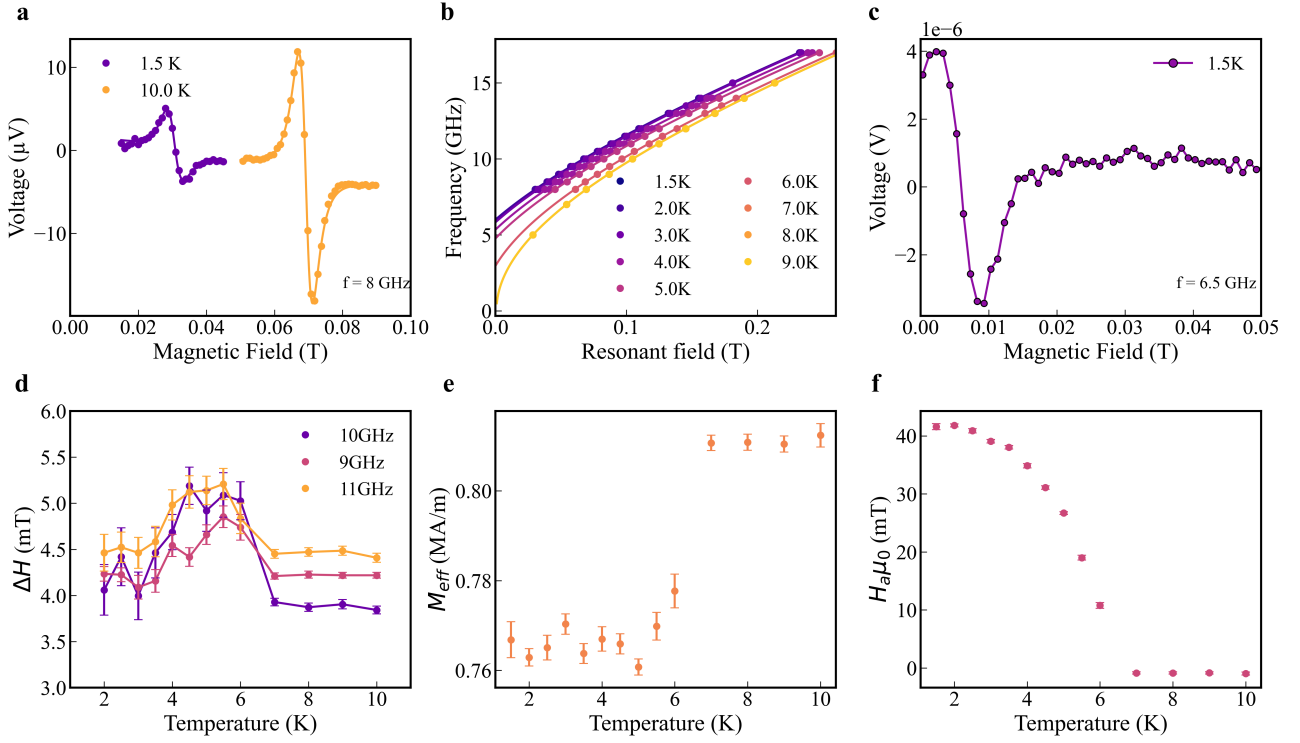


Figure 6: Temperature dependent ferromagnetic resonance (FMR) on Nb(80 nm)/Ni₈₀Fe₂₀(20 nm)/(Nb(80 nm)) across 5 - 18 GHz. (a) FMR absorption spectra at 8 GHz above T_c (10 K) and below T_c (1.5 K), showing a rigid 41 mT shift to lower fields and a decrease in signal below T_c . (b) Frequency resonant field relationship fitted with modified Kittel equation with the addition of rigid shift, showing a shift to lower fields. (c) At 1.5 K at 6.6 GHz there is a nonzero resonance at zero field. (d) FMR linewidth increases at T_c recovering at lower temperatures. (e) Effective magnetisation extracted from the modified Kittel fittings shows a drop reflecting a more negative second order perpendicular anisotropy. (f) Extracted anisotropy field showing an increase of 41 mT at 1.5 K demonstrating the superconducting shift.

Figure 6 displays the phenomenon observed with the shift of the FMR Lorentzians from above T_c , 10 K, and below T_c , 1.5 K at 41 mT at 8 GHz. The signal amplitude decreases drastically below T_c . This is due to the Meissner screening of the AC microwave field, which alters the magnetic field distribution felt by Ni₈₀Fe₂₀, leading to a weakening of FMR excitation. The full frequency and resonant field relationship is shown in Figure 6b. This is modelled by modifying the Kittel equation 25 with a rigid shift. This is shown in equation 52 where H_a is the anisotropy field produced by the superconductor and $M_{eff} = M + H_{2\perp}$ assuming easy axis is in-plane. The standard M_{eff} structure is $M_{eff} = M - H_{2\perp}$; however, due to $H_{2\perp}$ being negative, it can be written [11]:

$$f = \frac{\mu_0 \gamma}{2\pi} \sqrt{(H_{res} + H_a)(H_{res} + M_{eff} + H_a)}. \quad (52)$$

The frequency at which the system undergoes resonance at $H_{ext} = 0$ changes from 0 GHz to 6.5 GHz as the temperature is reduced from 9 K to 1.5 K. Figure 6 displays the essence of the phenomenon: a gradual shift to lower resonant field excitation as the temperature is reduced past the transition.

Figure 6c displays the intriguing result of the resonance occurring when $H_{ext}=0$, which means this additional field acting on the ferromagnet can act without any external stimulus. By extracting the linewidth from the plots at different frequencies, we can examine the damping and inhomogeneous line broadening in the system. The linewidth increases below T_c , which is expected from the literature and is linked to the onset of singlet superconductivity suppressing the outflow of spin currents since the Cooper pairs cannot transmit spin information, therefore reducing damping. However, the observation that the linewidth stabilises back the previous value above T_c has not been previously seen. Kun Rok et al. (2019) studied the same structures with a thick Nb layer of 100 nm and observed the same increase in linewidth below T_c and a slight dip below 2 K; however, this was not the same quantity as seen in this investigation [10]. Figure 6e shows the decrease in effective magnetisation, which corresponds to the second-order perpendicular anisotropy becoming more negative. This means that the hard axis is strengthening below T_c . Finally, the extracted rigid anisotropy field H_a is given in Figure 6 f, which shows the temperature variation of the anisotropy field produced by the superconductor in $Ni_{80}Fe_{20}$. The anisotropy field reaches 41 mT, which is a significant change. This suggests that the shift is temperature-dependent, and the effect strengthens as the superconducting state strengthens.

The damping parameters are extracted from the linewidth to determine the effect the superconductor has on the damping. This is shown in figure 7.

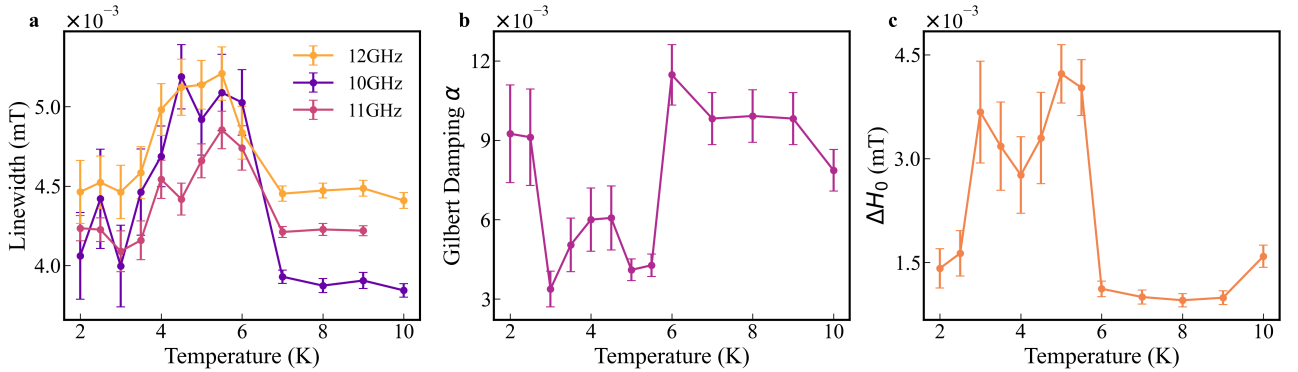


Figure 7: Damping parameters are extracted from the linewidth data shown in (a). (b) Gilbert damping decreases below T_c which is in line with literature [10]. (c) Inhomogeneous broadening increases below T_c which causes the increase in linewidth.

Figure 7 b shows the Gilbert damping as a function of temperature, and figure 7 c shows the inhomogeneous line broadening below T_c . The decrease in α seen below T_c is well documented in previous studies by Kun Rok et al., where they also saw an increase in ΔH_0 . This is due to the onset of the singlet superconducting state, which does not allow spin pumping out of Permalloy into Nb since the FMR energy is less than the superconducting gap; therefore, damping is reduced as spin cannot be lost [10]. The increase in Gilbert damping at 4K in figure 7 has not been seen before and the origin is unknown.

4.1 Asymmetric Structure

The next step in the investigation was to confirm that both superconducting layers are required to observe the increase in H_a . A sample of Nb110nm/Ni₈₀Fe₂₀20nm was studied using FMR to determine if there is any increase in anisotropy below T_c . Figure 8 displays the results.

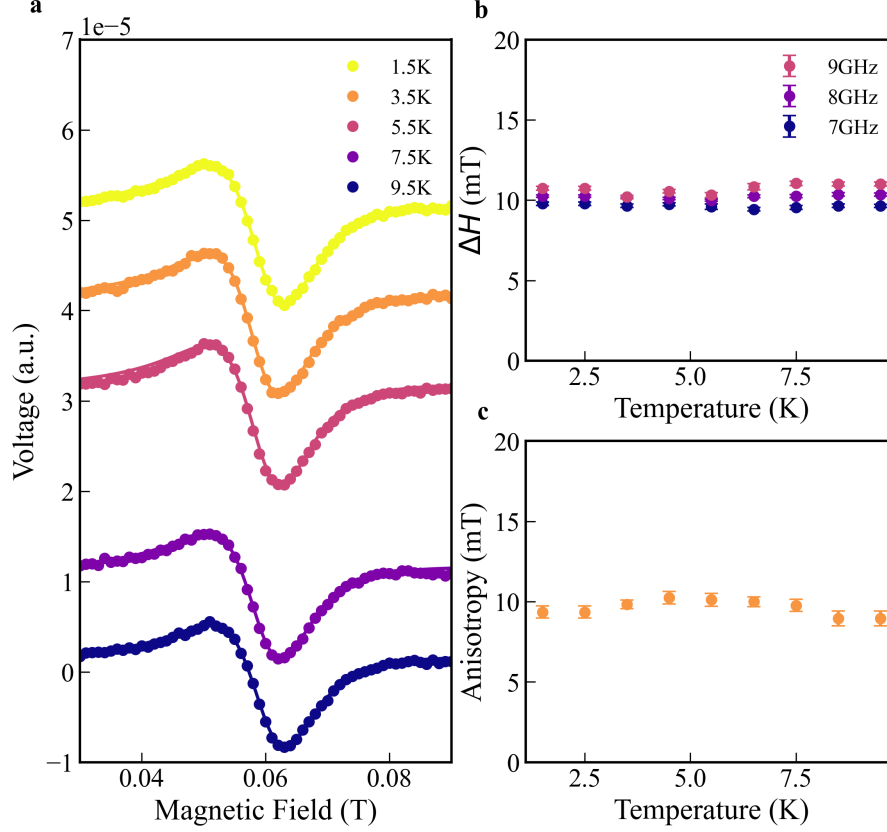


Figure 8: Asymmetric sample ferromagnetic resonance temperature dependence. (a) No shift seen through the superconducting transition. (b-c) This is reflected in the extracted linewidth and anisotropy since they remain constant.

Figure 8(a) shows the absorption curves for $f = 8$ GHz from 1.5K to 9.5K in 2K steps. From this figure, it is clear that there is no observed shift in the bilayer, indicating that two superconducting layers are required to observe this phenomenon. This is further reiterated in figure 8(a) and (c), where no noticeable change is observed in the anisotropy field data. This is consistent with previous studies, which also found that layers are required to produce this phenomenon [11, 10].

4.2 Thickness Dependence

The next crucial step is to determine the thickness dependence of the shift in H_r for both Nb and Ni₈₀Fe₂₀ thickness. In this experiment four samples were measured with S1 sample (Nb(110 nm)/Ni₈₀Fe₂₀(20 nm)/Nb(110 nm)) being a reference. Thicknesses of Nb were 80 nm, S4, and 50 nm, S5, and the thicknesses of Ni₈₀Fe₂₀ was 20 nm, S1, and 50 nm, S2, a further sample of Ni₈₀Fe₂₀ with

a thickness of 10 nm, S3, was grown ,however , the $\text{Ni}_{80}\text{Fe}_{20}$ thickness was too small to yield a good signal-to-noise ratio therefore was impossible to resolve below T_c . The results from this investigation are shown in figure 9.

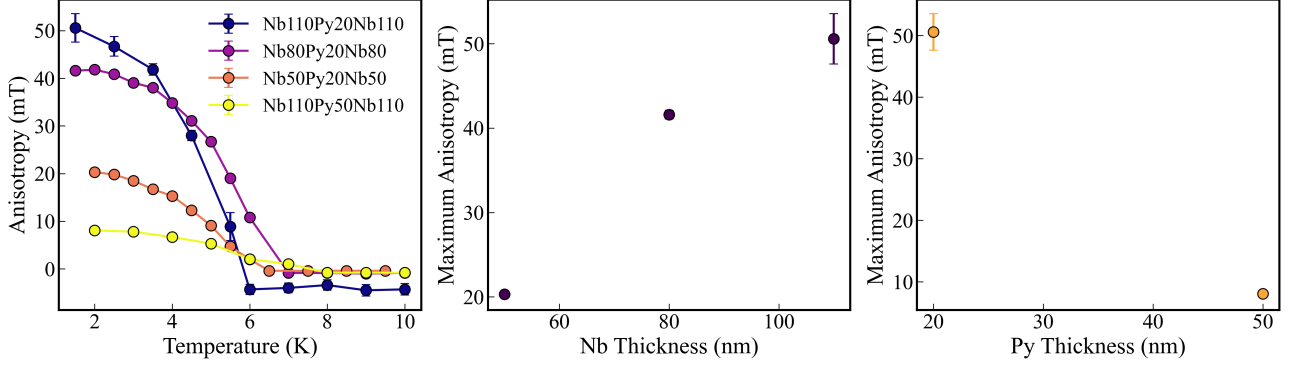


Figure 9: Thickness dependence of S/F/S by varying the thickness of the Nb layers and $\text{Ni}_{80}\text{Fe}_{20}$. (a) The extracted anisotropy field against temperature shows the effect is maximised for Nb(110 nm)/ $\text{Ni}_{80}\text{Fe}_{20}$ (20 nm)/Nb(110 nm) with a shift of 52 mT. (b) Increase in Nb thickness enhances the anisotropy shift. (c) Increase in $\text{Ni}_{80}\text{Fe}_{20}$ suggests a decrease in anisotropy however, this inconclusive.

Figure 9 (a) shows the extracted anisotropy fields from each sample as a function of temperature. The S1 sample exhibits the largest increase in the anisotropy field at 52 mT, which aligns well with previous studies [11] where they observed the maximum shift with thicker superconductor. This is followed by S4 with 41 mT, then S5 with 22 mT, and finally S3, which had a negligible increase of 8 mT. Figure 9 (b) shows the dependence of Nb thickness on the maximum anisotropy field observed, showing a positive correlation between Nb thickness and the increase in anisotropy. This is consistent with previous studies [11]. Figure 9 (c) may suggest that increasing $\text{Ni}_{80}\text{Fe}_{20}$ thickness, decreases the effect; however, since there is little data, this cannot be confirmed, and further samples must be measured to investigate this. This result also contradicts the literature, which has observed that increasing the $\text{Ni}_{80}\text{Fe}_{20}$ thickness enhances the effect. In the Golovchanskiy paper they suggest that the effect is due induced screening currents at the S/F interface which create a dynamic field acting on the magnetisation. Therefore, thicker permalloy increases the volume and moment of the precessing magnetisation which proportionally increases the induced superconducting currents. It is unclear why the results obtained in this investigation are different, however it could be due to issues with growth.

The increase in the effect with Nb thickness points towards the Meissner effect being the main contributor since if it was proximity-related, it would saturate at the penetration depth, which is typically 10-40nm for sputtered Nb [40]. Next, to determine whether it is a global Meissner effect or angle-dependent, measurements were taken.

5 Angular Dependence

The next step in this study is to determine the angular dependence of the H_r in order to see if there is a directional dependence. This investigation is important to determine if the main contribution to this effect is the Meissner effect, which should be isotropic in-plane. This will allow the anisotropy constants to be systematically mapped, thereby gaining a deeper understanding of the magnetic symmetry and the energy landscape.

In particular, the angular FMR analysis enables the identification of uniaxial and biaxial anisotropies. If the shift in H_r shows a significant angular dependence, it suggests that this could be due to stray field modifications, spin-triplet generation, interfacial orbital currents, or inverse proximity effects. Additionally, this approach enables us to distinguish between the intrinsic magnetic anisotropies of the ferromagnet and those induced by proximity to the superconducting layers.

5.1 In-Plane

In-plane measurements were taken in the zy plane to vary θ_H in 10° steps for 360° at 10 GHz, 12 GHz, and 14 GHz to investigate the frequency dependence in-plane. The S1 sample was used in this test since it showed the largest shift in H_r . The results are shown in figure 10.

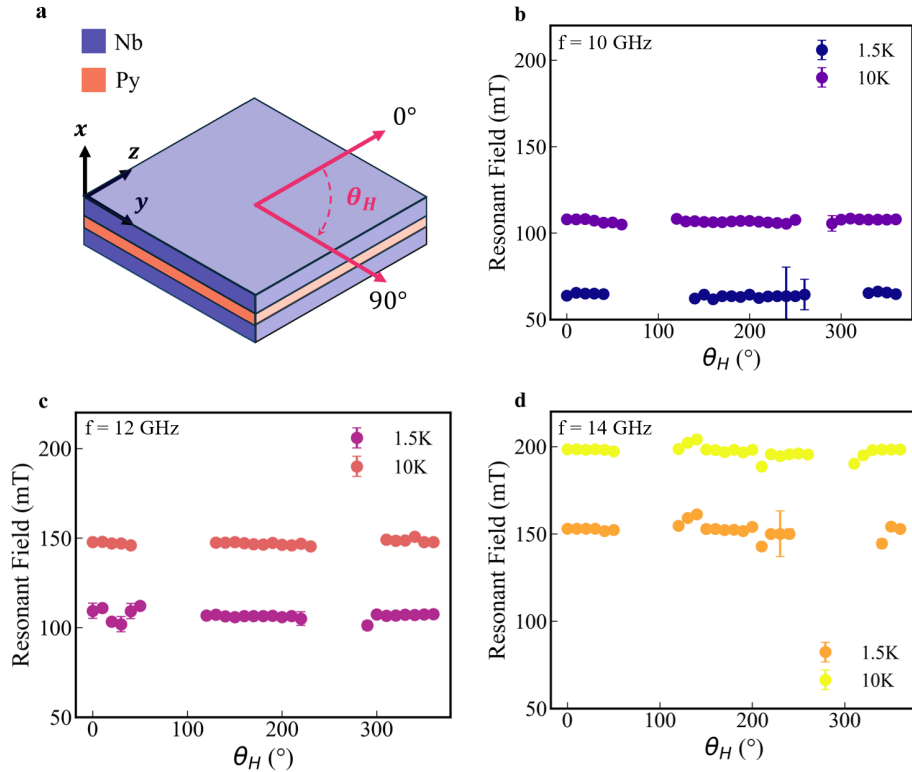


Figure 10: In-plane ferromagnetic resonance for different frequencies. (a) shows the coordinate system and the direction sweep. (b-d) Frequencies 10, 12 and 14 GHz are explored above and below T_c . The data suggests that the shift is isotropic in nature and frequency independent.

Figure 10(b-d) shows the variation of the H_r with respect to θ_H . From this, it is clear that the in-plane shift is isotropic, as there is negligible variation in the shift. The shift is also consistent across all frequencies, suggesting the rigid shift is frequency-independent. Since it is frequency-independent this goes against the proposed mechanism suggested by Silaev et al. which suggests a dynamical effect produced by the precession of the magnetisation [41]. This effect would be frequency dependent, however, this result points to a static effect.

The resonance was unable to be excited when moving towards 90° since the AC magnetic field produced by the microwave is parallel to the external field, causing little torque on the magnetisation; this, in turn, leads to inefficient excitation.

These results suggest that this is likely due to the Meissner effect or proximity effects, as these effects are angle-dependent and behave isotropically. These results rule out spin-orbit effects, such as the Rashba effect, since these would create a directional dependence. This is because Rashba spin orbit coupling links the electrons in-plane momentum to its spin which creates an effective field which rotates in-plane. As a result, rotating the magnetisation in-plane changes the relative orientation to the spins producing a directional dependence which would be measureable.

5.2 Out-of-Plane

To complete the picture of the anisotropy landscape plane, measurements were undertaken. Since the superconducting proximity effect, Meissner screening, or induced orbital currents may produce stray fields or anisotropic interactions perpendicular to the film plane, probing the resonance field's angular dependence out-of-plane allows us to detect such contributions. Ultimately, the out-of-plane data can provide insight into the magnetic energy landscape and help clarify the origin of the observed rigid resonance field shifts below T_c . For this investigation, S4 was studied due to its higher signal-to-noise ratio, which was crucial since moving out of the plane resulted in a smaller signal-to-noise ratio. This is due to limitations in the current setup, which only has field modulation in-plane, resulting in noise when the field is moved out of the plane. Consequently, it was difficult to resolve the absorption profile amidst the noise. Measurements were taken across the whole angle range in the zx plane by sweeping the angle φ_H in 5° increments. Figure 11 shows the results of the out-of-plane study.

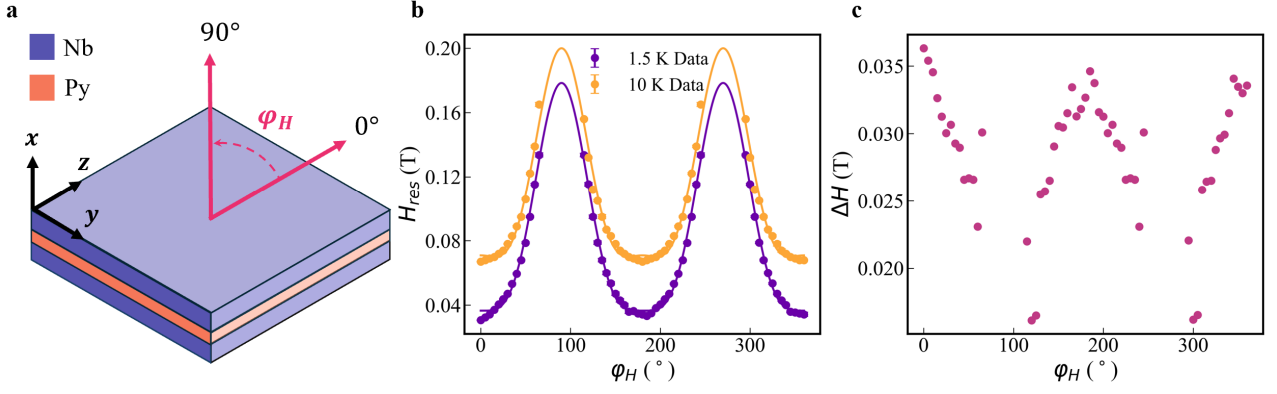


Figure 11: (a) Out-of-plane ferromagnetic resonance measurements sweeping the polar angle. (b) Above and below T_c show a uniaxial dependence with a in-plane easy axis. The data is fit with a phenomenological model to understand the changes in anisotropy. (c) The difference resonant field is not isotropic from above and below T_c suggesting a strengthening of the hard axis below T_c .

Figure 11 (b) shows the resonant field relationship with φ_H . It exhibits a cosine-squared relationship, which is consistent with the thin film geometry that creates a strong demagnetising field as magnetic poles form on the film surfaces, generating stray fields that increase the system's energy. Therefore, the system prefers in-plane magnetisation since this minimises energy. There is a clear shift between the 1.5K and 10K measurements; however, they are not isotropic. As φ_H goes towards the hard axis, there is less of a shift in the resonant field. The shift out-of-plane is not as strong, suggesting that it could be due to incomplete screening, resulting from geometric constraints in thin films, or that the superconductor does not want M to be out-of-plane.

To quantify the changes in the out-of-plane anisotropy landscape above and below T_c , a simple model has been used to fit the data and determine the effect that superconductivity has on the ferromagnet's anisotropy below T_c . The model is shown in equation 53, which is a simple model derived from linearising the resonance condition [25, 14].

$$H_{res} = H_0 + H_2 \cos 2\varphi_H + H_4 \cos 4\varphi_H \quad (53)$$

The fitting parameters H_2 are proportional to the two-fold anisotropy constant, which determines the easy or hard axis of the ferromagnet. H_4 is also included, as the system has weak four-fold anisotropy, which causes the peaks of the plot to be sharper. This is common in thin films. To reiterate, this is not a comprehensive treatment of free energy, but key information can still be extracted from this model. The offset is also a rigid shift, representing the angle-independent addition to the resonant field, which is the baseline field. This can also be affected by the onset of superconductivity. The results from fitting the data suggest a rigid shift of the resonant field by 28.3 mT to lower fields after the onset of superconductivity, which is similar to the in-plane data, indicating an angle-dependent shift in H_r . The magnitude of the shift is lower than previously recorded, and this is due to the change in anisotropy. This is reflected in the shift of the two fold anisotropy, which is proportional to the uniaxial

Parameter	10K (mT)	1.5K (mT)
H_0	113.1	84.8
H_2	-59.1	-65.4
H_4	14.0	12.9

Table 2: Fitting parameters for anisotropy constants.

anisotropy. In both the above and below T_c data, the H_2 parameter is negative, indicating an in-plane easy axis, which is expected for thin films due to the demagnetising field. The H_2 parameter becomes more negative below T_c , which suggests that the hard axis is strengthened below T_c , effectively making it even more energetically favourable to be in-plane. The hardening of the out-of-plane axis can be seen visibly from figure 11c as the shift is less pronounced out-of-plane, meaning the ferromagnet still requires the high field to orient out-of-plane higher than OOP if normalised with this rigid shift. This fitting is preliminary, and a full Smit Beljers numerical model will be developed to quantify the anisotropy constants accurately and confirm this conclusion [25].

6 Discussion

In this work, it has been demonstrated that in S/F/S structures below T_c , there is a rigid shift of the resonant field of the ferromagnet in response to the onset of superconductivity. This behaviour is in line with previous reports on these structures by Lai-Lai Li et al. (2018), Golovchanskiy et al. (2019), and Jeon (2019), who also observed the same effect [8, 11, 10]. The explanation from these papers was inconclusive, suggesting that it may be due to the Meissner effect, triplet generation, or magnetisation dynamics. The primary objective of this investigation thus far is to fully understand this effect through careful measurements across the superconducting transition.

In this study, a rigid shift of the resonant field is recorded. This has previously been attributed to a lensing effect of the external field, which causes the additional effective field felt by $\text{Ni}_{80}\text{Fe}_{20}$ [41]. However, the result in Figure 6 c shows a nonzero resonance at $H_{ext} = 0$ T, which suggests that it is independent of the external field. Additionally, since the Meissner effect is field-dependent, the Kittel relationship would be skewed by a field-dependent effective anisotropy field, which is not observed. This suggests that the Meissner effect in response to the external field is not possible. The next step in this investigation was to determine the angle dependence of the induced resonance shift by conducting angle-dependent FMR in-plane and out-of-plane measurements, which had not been previously studied. The in-plane results indicate that the shift in the resonant field is isotropic across all measured angles and frequencies. Therefore, the additional effective field produced by the superconductor always points in the direction of the $\text{Ni}_{80}\text{Fe}_{20}$ magnetisation. This result suggests that the origin is a lensing effect; however, it is not due to the external field but rather to the ferromagnet's magnetisation since the effect is not dependent on the external field. The ferromagnet produces its

own stray fields, which are, in turn, expelled by the superconductor. The symmetrical structure is required in this case since both interfaces of $\text{Ni}_{80}\text{Fe}_{20}$ have stray fields and require a superconductor on both sides to effectively expel the field into the ferromagnet creating flux pressure on $\text{Ni}_{80}\text{Fe}_{20}$. In the S/F, the stray fields just escape to the vacuum, and there is no pressure. The thickness dependence study, which shows that increasing the thickness of Nb increases the effect, is also in line with this interpretation, as the superconductor will have even stronger screening.

In terms of the out-of-plane results, they point towards a strengthening of the hard axis, which is also seen in the drop in M_{eff} below T_c . The out-of-plane configuration is not energetically favourable for the superconductor due to the nucleation of fluxes, which results in lowering the density of Cooper pairs since the normal state vortices penetrate more freely out of the plane. From the results so far, it is clear that the superconductor changes the internal energy of the ferromagnet in order to conserve the superconductor's internal energy. So, by introducing fluxes in the superconductor, there is a resultant increase in anisotropy in the ferromagnet, making it harder to bring the magnetisation out of the plane. This is why the hard axis hardens below T_c . This has been reported theoretically in a previous study by Aliev and Linder et al [42]. This could explain the strengthening of the hard axis below T_c .

Recently, a paper came out which added Pt, PMA, and Gd insertion-layer controls. These controls were to enhance triplet generation. However, the paper observed a decrease in shift in all these cases. Interestingly, shorting the Nb (sputtering on side walls) increased the effects which points further towards a lensing effect [43].

7 Future work

This investigation has laid the groundwork towards understanding the resonant shift in the resonance field below T_c of the superconductor in a S/F/S structure. The future work is outlined in four distinct parts, as shown below:

- Part 1:** Finalise understanding of the rigid shift in the resonant field below T_c in S/F/S structure.
- Part 2:** Determine whether the onset of superconductivity can reorient a ferromagnet in-plane with perpendicular magnetic anisotropy (PMA) below T_c in S/F/S.
- Part 3:** Explore inverted structures F/S/F to see if there is any superconductivity-induced exchange anisotropy.
- Part 4:** Develop a memory device in which the superconducting state switches a ferromagnet

7.1 Part 1: Finalising Understanding

To solidify our proposed mechanism for the rigid resonance-field shift, the thickness dependence study will be extended by exploring a wider range of $\text{Ni}_{80}\text{Fe}_{20}$ and Nb layer thicknesses. Varying the $\text{Ni}_{80}\text{Fe}_{20}$ thickness tests the idea that the shift's magnitude grows with the ferromagnet's available magnetic volume. Hence, we expect a larger rigid shift—up to the point where the screening is limited by the Nb penetration depth. Since in the limit of $\text{Ni}_{80}\text{Fe}_{20}$ thickness becoming infinity, the Nb will only affect the interface; therefore, the stack will act like two separate superconductors. Likewise, increasing the Nb thickness directly probes how the superconductor's Meissner screening length controls the effect. This will make the study more robust. This will also be supplemented by growing samples with copper(Cu) spacer layers between the $\text{Ni}_{80}\text{Fe}_{20}$ and Nb layers. The effect of adding spacers would be detrimental to the proposed mechanism since it would reduce the strength of the lensing effect. This is because the Cooper pair density is maximal at a clean S/F interface; however, since a Cu spacer is present, the Cooper pairs will leak in and decay, reducing the condensate away from the interface. Therefore, thicknesses such as 5 nm, 10 nm, 20 nm, and 40 nm will be studied, as the coherence length of Cu is typically around 30 to 50 nm when sputtered. The number of CPs interacting with $\text{Ni}_{80}\text{Fe}_{20}$ would scale with this thickness of Cu.

Secondly, a complete anisotropy model will be formed to fully extract the changes in anisotropy energy above and below T_c . The simple model gives us an idea of the behaviour; however, a rigorous model has a more physical basis. This will be done by solving the Sit Belijers equation for the out-of-plane case. First, the model will only contain uniaxial anisotropy, as this dominates, and it will be built from there. The initial idea is to use the same assumptions as mapped out in the derivation of the Kittel equation to form a macrospin model. However, if this is not enough, a complete numerical model will be formed which takes into account the variation between φ_H and φ_M .

They propose that the out-of-plane anisotropy is weakened due to the superconductor trying to conserve its free energy by reducing the amount of vortices. To verify that the increase in perpendicular anisotropy energy below T_c is compensated by condensation-energy loss due to vortex penetration. The change in anisotropy energy out-of-plane needs to be quantified via modelling, and then it can be directly compared to the reduction of the condensation energy from in-plane and out-of-plane. Quantifying the superconducting volume lost due to the formation of vortices can be done by performing a measurement of T_c with in-plane and out-of-plane magnetic field. This is to determine the change in T_c in both in-plane and out-of-plane directions. This can be directly compared to the loss of condensation energy since $u_{\text{cond}}(T) \approx \frac{1}{2} \mu_0 H_c^2(0)$ scales with $[1 - (T/T_c)^2]^2$. This change in energy can be compared to the increase in anisotropy energy. If the change in anisotropy energy is smaller than the change in condensation energy, the proposed mechanism is possible since it does not violate energy conservation [42].

Risk/Gain

This part of the project is relatively low risk, as the growth procedure and measurement technique have already been established. Therefore, it is only an additional task to the work outlined in this report. The modelling aspect should be straightforward since this theory is well understood and widely used in many FMR studies, and we are in contact with theorists who could provide guidance if needed. The comparison of energy contributions should be a straightforward test, as we have the facilities to perform T_c measurements at different angles in a magnetic field and extract the change in condensation energy. However, this relies on the creation of the model.

7.2 Part 2: Superconductor-Driven Reorientation of PMA Ferromagnets

From the results so far, it is clear that this effect enhances the anisotropy of the ferromagnet in-plane. This has been seen in $\text{Ni}_{80}\text{Fe}_{20}$ by the in-plane easy axis becoming easier below T_c . The next step is to try to harness this effect for switching applications, which requires changing the direction of the easy plane. Therefore, the next interesting step is to focus on ferromagnets with some perpendicular magnetic anisotropy (PMA). The issue with magnets with PMA is that they tend to be quite hard, which means the FMR measurements would be challenging since the resonance condition would shift to above 1T, making measurement impossible. Kun Rok et al. 2020 displayed that the PMA of Co can be tuned by having Pt/Co/Pt stacks since the interfacial spin-orbit coupling modifies the anisotropy landscape of Co. Kun Rok found that the optimum thickness of Co is 0.8 nm, which is very small in comparison to the thicknesses of $\text{Ni}_{80}\text{Fe}_{20}$ [44]. However, thicker layers also produce PMA, but not fully out-of-plane, but canted, which would be sufficient for this experiment. Another option is multilayers of Co/Pt or Co/Ni, where each layer contributes an interfacial anisotropy term [45]. In Beaujour et al. 2007, they performed FMR on Co/Ni multilayers of $n=10-20$. The PMA is maximised when the thickness of the layers is less than 0.6 nm [46].

To begin this investigation, inspiration from [46] will be explored since the volume of ferromagnet can be maximised in this way. Since they used Ni we will start with the $\text{Ni}_{80}\text{Fe}_{20}$ in a similar fashion. If this does not work Ni will be then used. First, the dependence of the layer thickness in both these ideas will be characterised to obtain the PMA of interest. Then, the structure of interest, as determined by the investigation, will be implemented into the S/F/S structure to determine if the mechanism proposed in this report can change the direction of the easy axis below T_c , thereby making an out-of-plane magnet in-plane.

Risk/Gain

This part of the project is more ambitious and inherently risky, as it has not been done before. However, previous studies on PMA using FMR have been successful. The growth of the PMA structure should be straightforward with the facilities available and local know-how since the Robinson group are experienced in this area. However, it will take time to develop the best structure for this specific

application since there were challenges with signal-to-noise ratio below T_c with the sample with 10 nm of $\text{Ni}_{80}\text{Fe}_{20}$. Therefore, maximising thickness of the ferromagnetic layer is important. However, with careful tuning of the thickness, this can be completed. To mitigate the risk, if this proves challenging, the project will shift towards inverted structures instead.

7.3 Part 3: Inverted Structure

By moving to an F/S/F geometry, the question becomes: Can the same Meissner-screening physics actually couple two separate ferromagnets through the superconducting spacer and thereby induce a superconductivity-mediated exchange anisotropy? An inverted structure will hence be fabricated and perform FMR on one layer by either using two separate magnetic materials, such as $\text{Ni}_{80}\text{Fe}_{20}$ and Co or by modulating the thickness of the two layers. By measuring the resonant field shift of the free F layer above and below T_c for each alignment, we can directly quantify any superconductivity-induced exchange anisotropy transmitted through the Nb spacer. A systematic, T_c -dependent difference in the resonance condition between parallel and antiparallel states would demonstrate that Meissner screening not only boosts a single F's anisotropy but also couples two F layers across the superconductor [47].

Risk/Gain

This part of the project is an addition to the S/F/S structure being studied and will be undertaken as a study if the PMA study proves too challenging within the time frame or if there is time for further investigation. This is risky as it has not been done before. However, there is likely interesting Physics which has not been explored in this area.

7.4 Part 4: Develop a Memory Device

Based on the design developed in parts 1-3 the main aim is to develop a memory device which utilises a superconductor which can switch a ferromagnet. This is the most ambitious part of the project as it relies on the previous parts.

7.5 Gantt Chart

Below in figure 12 is the proposed timeline for the rest of the project to ensure time is managed effectively to maximise my results in this project.

	Year 1			Year 2						Year 3					
Project Stage	June	August	Oct	Dec	Feb	April	June	August	Oct	Dec	Feb	April	June	August	Oct
Part 1															
Grow more thicknesses of Py															
Grow Cu spacers															
Anisotropy modelling															
Condensation energy experiment															
FMR measurement															
Part 2															
Design PMA structure															
Growth of different PMA ferromagnets															
Characterise PMA ferromagnets anisotropy															
Grow S/PMA/S															
FMR measurement															
Tune anisotropy with superconductor															
Part 3															
Grow F/S/F structures															
Characterise															
FMR measurements															
Thickness dependence growth															
FMR measurement															
Data analysis															
Thesis Writing															
Part 4:															

Figure 12: Timeline for the rest of PhD project taking into account the future work needed to produce good results.

8 Conclusion

This investigation aimed to investigate the resonant field shift of a ferromagnet in a superconductor/ferromagnet/superconductor structure below T_c , which corresponds to an effective increase in anisotropy that has been reported previously, yet the mechanism has been widely disputed. First, the work of Golovchanskiy et al. and others was reproduced by studying the shift in resonant field below T_c of different thicknesses of Nb and $\text{Ni}_{80}\text{Fe}_{20}$ [11]. The resonance behaviour was mapped for the full frequency field range. The second section of this investigation aimed to determine whether the shift in the resonant field is directional, which would indicate whether this is a global effect or not. This was done for both in-plane and out-of-plane.

In summary, the results demonstrate a temperature-dependent shift in the resonant field below T_c , reaching a significant value of 52 mT. This effect is independent of the external field as a nonzero resonance is seen at 0T. Control studies done on S/F bilayers show that this effect is only present in the symmetric structure. A systematic study of the thickness dependence of this effect shows that the thicker the Nb, the stronger the shift in the resonant field. The thickness dependence of the ferromagnetic layer suggests a reduction in the effect with thicker $\text{Ni}_{80}\text{Fe}_{20}$; however, this contradicts previous research and is inconclusive, as more thicknesses need to be tested. Previous studies have explained this effect in several ways. The first paper to report on this suggested that it was due to spin-triplet transfer torque on the ferromagnet; however, this structure is unlikely to produce long-range triplets. The second study on this effect pointed towards a dynamical effect that causes a gap to open at zero field, corresponding to Anderson-Higgs mass generation. However, the theoretical model presented in the corresponding theory paper does not fit the data exactly and assumes a thin Nb, which is not the case. Finally, other groups suggest that it is a Meissner screening effect of the effective field, which is

not possible since we observe zero-field resonance.

The second section focused on angle-dependent measurements to determine whether it is a global effect. The results indicate that the in-plane anisotropy is constant, resulting in an isotropic behaviour, and exhibits a consistent shift across all azimuthal angles. This points towards a global effect. The out-of-plane measurements show this shift to lower resonant frequencies; however, it is not isotropic in nature, as the shift is largest in-plane, and as M is rotated out-of-plane, the shift decreases. This points towards a hardening of the out-of-plane hard axis below T_c .

The current interpretation of this effect is that it is a lensing effect of the ferromagnet's own magnetisation. Previous studies suggested the Meissner effect; however, since we have concluded that there is a nonzero resonance at zero field, it cannot be due to the external field. This is a completely field-independent term since the Kittel plots show a rigid shift with no skewing, which would suggest a field-dependent component. This is further supported by the in-plane FMR study, which shows that the anisotropy is isotropic in-plane, consistent with the Meissner effect. The out-of-plane data is slightly more ambiguous; however, it can be explained by the superconductor causing the ferromagnet to resist the out-of-plane orientation in order to conserve the superconducting condensation energy as moving out-of-plane increases vortex nucleation, which reduces the number of Cooper pairs present.

To conclude this study, more measurements must be taken to prove further the hypothesis, such as adding spacer layers. To utilise this effect for the application of quantum memory storage, the idea is to move towards ferromagnets with perpendicular magnetic anisotropy. Since the superconductor wants the ferromagnets magnetisation to lie in-plane, it could be possible to utilise this to rotate the magnetisation of the PMA ferromagnet in-plane. This would show that a superconductor could be used to switch a ferromagnet, which is highly attractive for research in quantum memory storage.

References

- [1] Jacob Linder and Jason W. A. Robinson. Superconducting spintronics. *Nature Physics*, 11(4):307–315, 2015.
- [2] J. Bardeen, L. N. Cooper, and J. R. Schrieffer. Microscopic theory of superconductivity. *Physical Review*, 106:162–164, 1957.
- [3] E. Abrahams, A. Balatsky, D. J. Scalapino, and J. R. Schrieffer. Properties of odd-gap superconductors. *Physical Review B*, 52:1271–1278, 1995.
- [4] Ranran Cai, Igor Žutić, and Wei Han. Superconductor/ferromagnet heterostructures: A platform for superconducting spintronics and quantum computation. *Advanced Quantum Technologies*, 6(1):2200080, 2023.
- [5] VV Ryazanov, VA Oboznov, AS Prokofiev, VV Bolginov, and AK Feofanov. Superconductor—ferromagnet—superconductor π -junctions. *Journal of low temperature physics*, 136:385–400, 2004.
- [6] VV Ryazanov, VA Oboznov, A Yu Rusanov, AV Veretennikov, Alexandre Avraamovitch Golubov, and J Aarts. Coupling of two superconductors through a ferromagnet: Evidence for a π junction. *Physical review letters*, 86(11):2427, 2001.
- [7] AK Feofanov, VA Oboznov, VV Bol’Ginov, J Lisenfeld, S Poletto, VV Ryazanov, AN Rossolenko, M Khabipov, D Balashov, AB Zorin, et al. Implementation of superconductor/ferromagnet/superconductor π -shifters in superconducting digital and quantum circuits. *Nature Physics*, 6(8):593–597, 2010.
- [8] Lai-Lai Li, Yue-Lei Zhao, Xi-Xiang Zhang, and Young Sun. Possible evidence for spin-transfer torque induced by spin-triplet supercurrents. *Chinese Physics Letters*, 35(7):077401, 2018.
- [9] Manuel Houzet. Ferromagnetic josephson junction with precessing magnetization. *Physical review letters*, 101(5):057009, 2008.
- [10] Kun-Rok Jeon, Chiara Ciccarelli, Hidekazu Kurebayashi, Lesley F Cohen, Xavier Montiel, Matthias Eschrig, Thomas Wagner, Sachio Komori, Anand Srivastava, Jason WA Robinson, et al. Effect of meissner screening and trapped magnetic flux on magnetization dynamics in thick nb/ni 80 fe 20/nb trilayers. *Physical Review Applied*, 11(1):014061, 2019.
- [11] IA Golovchanskiy, NN Abramov, VS Stolyarov, VI Chichkov, M Silaev, IV Shchetinin, AA Golubov, VV Ryazanov, AV Ustinov, and M Yu Kupriyanov. Magnetization dynamics in proximity-coupled superconductor-ferromagnet-superconductor multilayers. *Physical review applied*, 14(2):024086, 2020.

- [12] IA Golovchanskiy, NN Abramov, OV Emelyanova, IV Shchetinin, VV Ryazanov, AA Golubov, and VS Stolyarov. Magnetization dynamics in proximity-coupled superconductor-ferromagnet-superconductor multilayers. ii. thickness dependence of the superconducting torque. *Physical Review Applied*, 19(3):034025, 2023.
- [13] Stephen Blundell. *Magnetism in condensed matter*. OUP Oxford, 2001.
- [14] Nicola A Spaldin. *Magnetic materials: fundamentals and applications*. Cambridge university press, 2010.
- [15] Zhou Fang. *Current-induced torque in ferromagnets at room temperature*. Phd thesis, University of Cambridge, Jesus College, 2017.
- [16] David J. Griffiths. *Introduction to Quantum Mechanics*. Pearson Prentice Hall, 2nd edition, 2005.
- [17] J. J. Sakurai and Jim Napolitano. *Modern Quantum Mechanics*. Addison-Wesley, 2nd edition, 2010.
- [18] W. Heisenberg. Zur theorie des ferromagnetismus. *Zeitschrift für Physik*, 49:619–636, 1928.
- [19] N. W. Ashcroft and N. D. Mermin. *Solid State Physics*. Holt, Rinehart and Winston, 1976.
- [20] J. H. Van Vleck. *The Theory of Electric and Magnetic Susceptibilities*. Oxford University Press, 1932.
- [21] Charles Kittel. *Introduction to Solid State Physics*. Wiley, 8th edition, 2005.
- [22] LALE Landau, Evgeny Lifshitz, et al. On the theory of the dispersion of magnetic permeability in ferromagnetic bodies. *Phys. Z. Sowjetunion*, 8(153):101–114, 1935.
- [23] Thomas L Gilbert. A lagrangian formulation of the gyromagnetic equation of the magnetization field. *Phys. Rev.*, 100:1243, 1955.
- [24] Bretislav Heinrich and J Anthony C Bland. *Ultrathin magnetic structures II: Measurement techniques and novel magnetic properties*, volume 2. Springer Science & Business Media, 2005.
- [25] Xinyu Liu and Jacek K Furdyna. Ferromagnetic resonance in $\text{Ga}_{1-x}\text{Mn}_x\text{As}$ dilute magnetic semiconductors. *Journal of Physics: Condensed Matter*, 18(13):R245, 2006.
- [26] J Gil-Monsalve, JE Abrão, E Santos, A Ricalde, A Azevedo, and O Arnache. Twofold and fourfold anisotropies in zinc ferrite thin films investigated by ferromagnetic resonance. *Physical Review B*, 105(1):014420, 2022.
- [27] Charles Kittel. On the theory of ferromagnetic resonance absorption. *Physical review*, 73(2):155, 1948.
- [28] Michael Tinkham. *Introduction to Superconductivity*. Dover Publications, 2nd edition, 1996.

- [29] David R Tilley. *Superfluidity and superconductivity*. Routledge, 2019.
- [30] James F Annett. *Superconductivity, superfluids and condensates*, volume 5. Oxford University Press, 2004.
- [31] VL Ginzburg. Zh. vl ginzburg and ld landau. *Zh. Eksp. Teor. Fiz*, 20:1064, 1950.
- [32] Physics Vidya Pith. Superconductors and its properties, February 2024.
- [33] Alexandre I Buzdin. Proximity effects in superconductor-ferromagnet heterostructures. *Reviews of modern physics*, 77(3):935–976, 2005.
- [34] Alexandre I Buzdin, AS Mel’Nikov, and NG Pugach. Domain walls and long-range triplet correlations in sfs josephson junctions. *Physical Review B—Condensed Matter and Materials Physics*, 83(14):144515, 2011.
- [35] FS Bergeret, AF Volkov, and KB Efetov. Long-range proximity effects in superconductor-ferromagnet structures. *Physical review letters*, 86(18):4096, 2001.
- [36] Peter J Kelly and R Derek Arnell. Magnetron sputtering: a review of recent developments and applications. *Vacuum*, 56(3):159–172, 2000.
- [37] Oxford Instruments Nanoscience. Teslatronpt dry cryogenic system. <https://nanoscience.oxinst.com/dry-systems/products/teslatronpt>, 2025. Accessed: 2025-06-30.
- [38] David M Pozar. *Microwave engineering: theory and techniques*. John wiley & sons, 2021.
- [39] Alex K Chan, Murat Cubukcu, Xavier Montiel, Sachio Komori, Alexander Vanstone, Juliet E Thompson, Garry K Perkins, CJ Kinane, AJ Caruana, D Boldrin, et al. Controlling spin pumping into superconducting nb by proximity-induced spin-triplet cooper pairs. *Communications Physics*, 6(1):287, 2023.
- [40] AI Gubin, KS Il’in, SA Vitusevich, M Siegel, and N Klein. Dependence of magnetic penetration depth on the thickness of superconducting nb thin films. *Physical Review B—Condensed Matter and Materials Physics*, 72(6):064503, 2005.
- [41] Mikhail Silaev. Anderson-higgs mass of magnons in superconductor-ferromagnet-superconductor systems. *Physical Review Applied*, 18(6):L061004, 2022.
- [42] César González-Ruano, Lina G Johnsen, Diego Caso, Coriolan Tiusan, Michel Hehn, Niladri Banerjee, Jacob Linder, and Farkhad G Aliev. Superconductivity-induced change in magnetic anisotropy in epitaxial ferromagnet-superconductor hybrids with spin-orbit interaction. *Physical Review B*, 102(2):020405, 2020.
- [43] Jiaxu Li, Jie Zhang, Guang Yang, and Weisheng Zhao. Efficient shift of ferromagnetic resonance by superconductor gating. *Applied Physics Reviews*, 12(1), 2025.

- [44] Kun-Rok Jeon, Xavier Montiel, Sachio Komori, Chiara Ciccarelli, James Haigh, Hidekazu Kurebayashi, Lesley F Cohen, Alex K Chan, Kilian D Stenning, Chang-Min Lee, et al. Tunable pure spin supercurrents and the demonstration of their gateability in a spin-wave device. *Physical Review X*, 10(3):031020, 2020.
- [45] William K Peria, D-L Zhang, Yihong Fan, J-P Wang, and Paul A Crowell. Anomalous temperature dependence of phonon pumping by ferromagnetic resonance in co/pd multilayers with perpendicular anisotropy. *Physical Review B*, 106(6):L060405, 2022.
- [46] J-ML Beaujour, W Chen, K Krycka, C-C Kao, JZ Sun, and AD Kent. Ferromagnetic resonance study of sputtered co—ni multilayers. *The European Physical Journal B*, 59:475–483, 2007.
- [47] IF Lyuksyutov* and VL Pokrovsky. Ferromagnet–superconductor hybrids. *Advances in Physics*, 54(1):67–136, 2005.